# SITE-SAFE: A DIGITAL ECOSYSTEM FOR AUTONOMOUS CONSTRUCTION SAFETY

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# ABSTRACT

Construction sites need evidence that safety controls were followed, yet manual inspections and passive CCTV provide only intermittent oversight. Site-Safe is a camera-first safety compliance system that detects hazards from site cameras, identifies the workers involved, alerts supervisors on web and mobile clients in real time and keeps an auditable per-worker record. This report describes its design, implementation and evaluation, and separates what was measured from what remains to be verified.

The implemented system combines a ten-class YOLO11-medium PPE detector, QR badge identification of workers, pose-based fall detection, image-plane vehicle proximity warnings, camera inactivity tracking and a fire and smoke stage whose severity follows hot-work permits. Camera events are buffered in a local outbox and posted with per-camera device tokens to a Django REST and WebSocket backend on PostgreSQL, which provides role-based access control, configurable alert rules, push and e-mail notification, automatic attendance and idle timelines, and DOCX and PDF reports. A React command centre with a site map and a React Native (Expo) mobile application are the clients. The Smart Vest's software path is implemented and exercised with a labelled simulator; the physical vest was moved to future work.

All 318 automated tests pass. On a single Apple M4 host, events reached the dashboard with a 95th-percentile latency of 42 ms, all 100 events queued during a five-minute outage were stored exactly once about 30 s after the API returned, and the full detector set ran at 15.7 frames per second on the laptop GPU. The PPE model measured mAP@0.5 of 0.562 on the held-out split, but missing-equipment classes remain weak (no-gloves 0.062, no-boots 0.130) and the published split places near-duplicate images in both training and test data, so the score is an upper bound. Fire detection awaits a trained model, and the two-hour soak test, physical QR range tests and push delivery to a phone are outstanding.

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# LIST OF ABBREVIATIONS

**API** Application Programming Interface

**ASGI** Asynchronous Server Gateway Interface

**CI** Continuous Integration

**DFD** Data Flow Diagram

**ERD** Entity Relationship Diagram

**FCM** Firebase Cloud Messaging

**FPS** Frames Per Second

**FR** Functional Requirement

**GPS** Global Positioning System

**IoT** Internet of Things

**JWT** JSON Web Token

**LoRaWAN** Long Range Wide Area Network

**mAP** Mean Average Precision

**MQTT** Message Queuing Telemetry Transport

**NFR** Non-Functional Requirement

**NMS** Non-Maximum Suppression

**PPE** Personal Protective Equipment

**QR** Quick Response (code)

**REST** Representational State Transfer

**RTSP** Real-Time Streaming Protocol

**SDG** Sustainable Development Goal

**YOLO** You Only Look Once

# Chapter 1

# Introduction

## 1.1 Background

Walk onto any large construction site in Pakistan and the scene is immediately familiar: dozens of workers moving across multiple floors, heavy machinery operating within close proximity to pedestrians, narrow scaffolding above unprotected drops, and perhaps one or two safety officers trying to keep track of it all. The practical impossibility of this arrangement is obvious to anyone who sees it. What is less obvious is how consistently this arrangement fails, and how predictably those failures result in death or serious injury.

According to the International Labour Organization, construction accounts for roughly 30 to 40 percent of all fatal occupational accidents worldwide despite employing only about 7 percent of the global workforce [16]. In Pakistan, the Labour Force Survey 2020–21 recorded over 3.6 million workers employed in the construction sector, most of whom receive little to no formal safety training and work on sites where PPE compliance is enforced, if at all, through periodic manual checks [17]. Projects funded through initiatives like CPEC have dramatically increased the volume and scale of construction activity without a corresponding investment in safety infrastructure.

The standard approach to construction safety, assigning a safety officer to patrol a site, look for violations, and respond to incidents, has not meaningfully changed in decades. It is a model that was workable when sites were small and crews were compact. On modern sites spanning multiple floors and hundreds of metres, it is not. A single officer physically cannot watch fifty workers simultaneously. Environmental hazards like heat buildup or gas accumulation in excavation zones are invisible to the naked eye. And when something does go wrong, the response is almost always reactive: help arrives after the fall, after the gas exposure, after the equipment strike.

The technological tools to do better than this now exist. Deep learning models can process camera feeds and detect safety violations in real time. Simple printed QR codes can identify individual workers without requiring any electronics on the person. Cross-platform mobile frameworks allow supervisors to receive alerts and monitor workers from anywhere on the site through their phones. When combined, these technologies can create something that has never existed on most Pakistani construction sites: an automated, per-worker compliance record that documents who was wearing what, where, and when. What has been missing is a system that brings all of these capabilities together into a coherent, practical package designed for actual deployment. That is what Site-Safe is intended to be.

## 1.2 Problem Statement

The specific problem this project addresses is the absence of a continuous, autonomous safety compliance system for construction sites—one that can detect hazards in real time, identify which worker is involved, alert relevant personnel immediately through both web and mobile interfaces, and maintain a verifiable, per-worker record of safety compliance without depending entirely on human attention.

The deficiencies in the current state of the art can be summarized under four headings. First, incomplete coverage: no human supervisor can simultaneously monitor all workers across a multi-floor or multi-zone construction site, and violations happen most frequently precisely when no one is watching. Second, purely reactive response: existing systems only register an incident after it has occurred, with no mechanism for detecting warning signs of an impending accident. Third, no individual accountability: even when a violation is observed, there is no automated way to identify which worker was involved or track repeat offenders across shifts. Fourth, no integrated compliance record: current sites maintain no systematic digital record of PPE compliance, worker activity patterns, or incident history tied to specific individuals, preventing any data-driven safety improvement or auditable compliance documentation over time.

## 1.3 Objectives

The work carried out under this project is guided by the following specific objectives:

Design and train a real-time PPE detection model based on YOLOv11 that can identify safety equipment across ten detection classes (five items of equipment, each worn and missing) and flag violations under the varied lighting and environmental conditions of an actual construction site.

Implement a QR-based worker identification system where each worker is assigned a unique QR code printed on their helmet, readable by the same cameras running PPE detection, enabling per-worker violation tracking and compliance accountability.

Implement a virtual geofencing system that allows administrators to mark danger zones on a site map, with automatic alerts when workers enter restricted areas, linked to individual worker identity through QR identification.

Develop a web-based command center that aggregates all incoming data and presents it to site managers through an interactive real-time dashboard.

Build a cross-platform mobile application (React Native) that enables safety officers to receive push notifications for critical alerts, view live worker status, and allows administrators to configure and manage alert settings remotely.

Implement workforce activity monitoring through the vision pipeline, tracking worker presence across zones, detecting extended breaks and inactivity periods, and generating per-worker activity logs.

Establish automated incident logging and PDF compliance report generation that documents per-worker violation histories, zone attendance records, and shift summaries, providing construction firms with auditable safety documentation.

Build a lightweight IoT Smart Vest with GPS tracking and an SOS panic button to provide supplementary worker location data and emergency alerts in areas outside camera coverage. The vest's software path is delivered and tested with a simulator; the physical hardware was moved to future work through a scope-change request to the supervisor (Section 4.2.2).

Integrate camera-based fall detection using YOLOv11-Pose, which analyses body keypoints frame-by-frame to identify when a worker has collapsed. A three-second persistence filter reduces false positives from bending or crouching, enabling the system to detect genuine falls and dispatch critical alerts without any wearable sensors.

Deploy a fire and smoke detection model running alongside the PPE pipeline on IP camera feeds. The module includes zone-aware suppression logic so that scheduled hot-work activities such as welding or cutting do not trigger false alarms, while genuine uncontrolled fires still generate critical alerts.

## 1.4 Scope

This project covers the full-stack design and prototype implementation of Site-Safe. The primary focus is software: the YOLOv11-based vision pipeline with a QR code reader for worker identification, a pose-estimation module for camera-based fall detection, fire and smoke detection governed by hot-work permits, the Django REST backend, the React web dashboard, a cross-platform React Native mobile application for real-time alert management and worker monitoring, and automated compliance reports. The software path of the supplementary Smart Vest (telemetry API, MQTT consumer and a labelled

simulator) is in scope; the physical vest hardware was moved to future work through a scope-change request to the supervisor (Section 4.2.2).

The software scope explicitly includes workforce activity monitoring, compliance documentation, fall detection, and fire safety features. The vision pipeline records when identified workers are on site and when they are idle, producing queryable activity timelines tied to individual worker identities via QR codes. A parallel pose-estimation model monitors body keypoints to detect falls, while a dedicated fire and smoke stage provides early fire warning whose severity is lowered, but never suppressed, during permitted hot-work activities. The mobile application gives supervisors and administrators remote access to alerts and worker status. Automated compliance reports document per-worker violation histories and attendance records.

What is explicitly outside the current scope: the system is designed for outdoor civil construction sites, not enclosed factory floors, mining tunnels, or marine environments. QR code identification requires workers to be within readable range of a camera, so the system works best at key checkpoints and designated work zones rather than across every square metre of a large open site.

## 1.5 Significance

The relevance of this project goes beyond the technical. Every year, preventable accidents on Pakistani construction sites cut short lives that did not need to be lost. Site-Safe is built on the premise that the technology to prevent many of those accidents already exists; the gap is in deployment, integration, and accessibility.

From a regulatory and commercial standpoint, construction firms face growing legal liability for workplace accidents. A system that automatically logs compliance evidence and generates auditable reports has direct economic value to any firm operating at scale. The mobile application component ensures that safety monitoring is not confined to a control room but extends to every supervisor carrying a smartphone on site.

## 1.6 SDG Alignment

SDG 3 (Good Health and Well-being): Real-time PPE violation detection, automatic fall detection through pose estimation, and fire and smoke monitoring enable timely intervention before unsafe behaviour leads to injury, directly reducing preventable construction fatalities.

SDG 8 (Decent Work and Economic Growth): Systematically safer working conditions for construction workers, regardless of skill level or contract status, align directly with the decent work agenda. The activity monitoring features add transparency to working hour tracking.

SDG 9 (Industry, Innovation, and Infrastructure): The application of AI, IoT, and mobile technology to modernize construction safety infrastructure is a concrete demonstration of the technology-driven industrial innovation SDG 9 envisions.

SDG 11 (Sustainable Cities and Communities): The fire and smoke detection module, with hot-work permits that distinguish scheduled flame work from uncontrolled fires, contributes to safer urban construction environments, reducing the risk of uncontrolled fires on building sites that endanger both workers and surrounding communities.

## 1.7 Report Organization

Chapter 2 reviews existing systems and relevant published research, identifying the specific gap that Site-Safe fills. Chapter 3 describes the proposed system design: architecture, module specifications including QR-based identification, fall detection, fire and smoke detection, workforce monitoring, mobile application design, and data flow. Chapter 4 describes the implementation: the development environment, QR identification and the Smart Vest software path, the vision modules, the backend, web and mobile applications, workforce activity monitoring, and the component, deployment, data and interaction views of the delivered system. Chapter 5 reports testing against the requirements traceability matrix, model evaluation and measured performance and reliability. Chapter 6 compares the outcomes with the objectives and sets out limitations and future work.

# Chapter 2

# Review of Existing Systems

## 2.1 Overview

Before settling on the approach described in Chapter 3, we surveyed the landscape of existing solutions, both commercial products and academic research, that attempt to address construction site safety. What this review revealed, fairly consistently, is that every available solution addresses one piece of the problem while leaving the others untouched. The following sections document what we found.

## 2.2 Traditional CCTV Surveillance

CCTV remains the most widely used safety monitoring technology in construction environments and has been for decades. Networks of fixed cameras feed into digital recorders accessible via monitor banks or remote connections. In their original form, these systems are entirely passive: they record what happens without analyzing it.

The fundamental limitation of CCTV is not the camera; it is the human being required to watch it. Sustained attention is cognitively demanding, and studies have shown that human observers miss a significant proportion of critical events when monitoring multiple camera feeds over extended periods. Sacks et al. (2022) noted that maintaining basic coverage on a medium-sized construction site required approximately 4.2 full-time monitoring personnel, a cost few firms can justify [6]. Beyond the attention problem, CCTV gives no information about what a camera cannot see: gas concentration in a trench, temperature inside a poorly ventilated enclosure, or how long a worker has been stationary in a hazardous zone. Its role in Site-Safe is not eliminated but redefined: cameras become the front end of an AI inference and activity tracking pipeline, not passive recording devices.

## 2.3 Consumer Smartwatches and Fitness Wearables

The Apple Watch and comparable devices from Samsung and Garmin have introduced genuinely impressive sensing capabilities into a wrist-worn form factor. Modern smartwatches can detect falls, monitor heart rate, flag irregular cardiac rhythms, and trigger emergency SOS calls. For personal health use, they are excellent.

For construction deployment, they fall short in several practical ways. Bluetooth and Wi-Fi connectivity work well in homes and offices; on a large outdoor site, they become unreliable as soon as a worker moves beyond router range. Battery life of 24 to 48 hours sounds adequate but does not account for continuous sensor polling under the

computational load of real-time monitoring. And devices designed for a commuter are not designed to survive the concrete dust, water exposure, and physical impact of a construction shift. At a retail cost of PKR 60,000 to 150,000 per unit, equipping an entire site crew is financially impractical for the vast majority of Pakistani construction firms.

## 2.4 Specialized Industrial Wearables

SmartCap Technologies produces a helmet-mounted EEG sensor designed to detect driver fatigue by reading brainwave activity. Within its target application, long-haul trucking and heavy vehicle operation in mining, it performs well. Outside that target, it offers nothing: no PPE compliance detection, no environmental gas sensing, no fall detection capability, and no location or health data beyond fatigue state.

This pattern, high accuracy within a narrow domain and zero applicability outside it, repeats across the industrial wearables category. Proximity warning tags alert workers when machinery approaches but do not monitor heat stress. Heat stress monitors do not detect falls. Fall detectors do not watch for gas. The fragmentation reflects the fact that no existing product was designed with the multi-hazard complexity of a construction site as its primary design constraint.

## 2.5 AI-Based PPE Detection Research

The academic literature on computer vision applied to construction safety is substantial and growing. The YOLO family of models, introduced by Redmon et al. in 2016, refined in YOLOv3 [1] and developed through successive iterations to YOLOv11 [3], has become the standard architectural choice for real-time PPE detection. The single-pass inference design achieves detection speeds and accuracy levels that are genuinely usable in operational contexts.

Published results are encouraging. Nath, Behzadan, and Paal (2020) achieved 91.2% accuracy on a construction site dataset using YOLOv3 [5]. More recent work with YOLOv8 and YOLOv11 pushes mean Average Precision above 92% on standard benchmarks. The problem is that virtually none of this research moves beyond the dataset. The overwhelming majority of published PPE detection studies evaluate their models on pre-recorded video in controlled settings. They do not deploy on actual edge hardware, integrate with sensor networks, generate real-time alerts, or connect to mobile applications.

They are demonstrations of what is possible in software, not implementations of what works in the field.

## 2.6 Integrated IoT Safety Platforms

A small number of research prototypes and commercial platforms have attempted to combine sensing modalities for construction safety. Ramli et al. (2020) published a notable IoT-based smart helmet prototype incorporating temperature, gas, and accelerometer sensors with GSM communication [4]. It was a meaningful step, but GSM communication provides only short-range connectivity, the prototype lacks any vision component, and there is no centralized data platform.

Commercial platforms like Pillar Technologies and Triax Technologies offer mesh-networked environmental sensor nodes and BLE worker tags for large construction sites. They provide real data and real utility. They also, notably, do not include computer vision, do not offer long-range communication for large open sites, and are priced for large Western construction enterprises rather than the Pakistani market.

## 2.7 Mobile Safety Applications

A growing category of construction safety tools focuses on mobile applications. Apps like iAuditor (now SafetyCulture) and Safesite provide digital checklists, incident reporting forms, and basic alert notifications. These tools digitize the paperwork side of safety management and make compliance documentation more convenient.

However, they remain fundamentally manual: a human must observe a violation, open the app, and log it. They do not detect hazards autonomously, do not integrate with IoT sensors or cameras, and do not provide real-time monitoring of worker vitals or location. They are essentially digital clipboards. Site-Safe takes the mobile component further: the mobile app is not just a reporting tool but a real-time monitoring interface connected to an autonomous detection backend, receiving push alerts the instant a hazard is detected and enabling administrators to configure the entire alert system remotely.

## 2.8 Comparative Summary

***Table 2.1:*** *Comparison of Existing Systems/Devices Against Site-Safe*

| System/Device | Year | Vision (AI) | Worker ID / IoT | Long-Range | Mobile App | Dashboard | Primary Limitation |
|---|---|---|---|---|---|---|---|
| Traditional CCTV | N/A | No | No | No | No | No | Passive, no analytics |
| Apple/Samsung Watch | 2023 | No | Partial | No | Yes | No | Short range, expensive |
| SmartCap (Mining) | 2019 | No | EEG only | No | No | No | Single-purpose |
| YOLO Research | 2024 | Yes | No | No | No | No | Software-only, no deploy |
| Pillar Technologies | 2022 | No | Yes | BLE | Partial | Partial | No vision, enterprise price |
| SafetyCulture App | 2023 | No | No | N/A | Yes | No | Manual, no detection |
| Ramli et al. | 2020 | No | Yes | GSM | No | No | No vision, no platform |
| Site-Safe (ours) | 2026 | Yes | QR (+ GPS vest, proposed) | LoRa (proposed) | Yes | Yes | Prototype; QR range not yet measured |

The table above makes the gap plain. No existing system, commercial or academic, combines computer vision-based PPE detection with QR-based individual worker identification, workforce activity monitoring, automated compliance documentation, a centralized web dashboard, and a mobile application for real-time alerts and administration. Site-Safe is designed to be that system.

# Chapter 3

# Proposed System Design

## 3.1 System Overview

Site-Safe is structured around three distinct layers that work together: edge devices on the construction site that capture and process visual data, a network layer that moves that data reliably from site to server, and an application layer that stores, processes, displays information through a web dashboard, and pushes critical alerts to a mobile application. The design deliberately separates these concerns so that individual components can be upgraded or replaced without requiring changes to the whole system.

On the site itself, the primary edge devices are AI cameras running YOLOv11 inference for both PPE detection and QR-based worker identification. Each worker wears a helmet with a unique printed QR code, which the same camera reads to establish identity without requiring any electronics on the person. A lightweight Smart Vest with GPS and SOS capability is proposed for supplementary tracking in areas outside camera coverage (Section 4.2.2). Everything converges at the backend server, which processes incoming data, evaluates alert rules, generates compliance records, and pushes updates to both the web dashboard and the mobile app in real time.

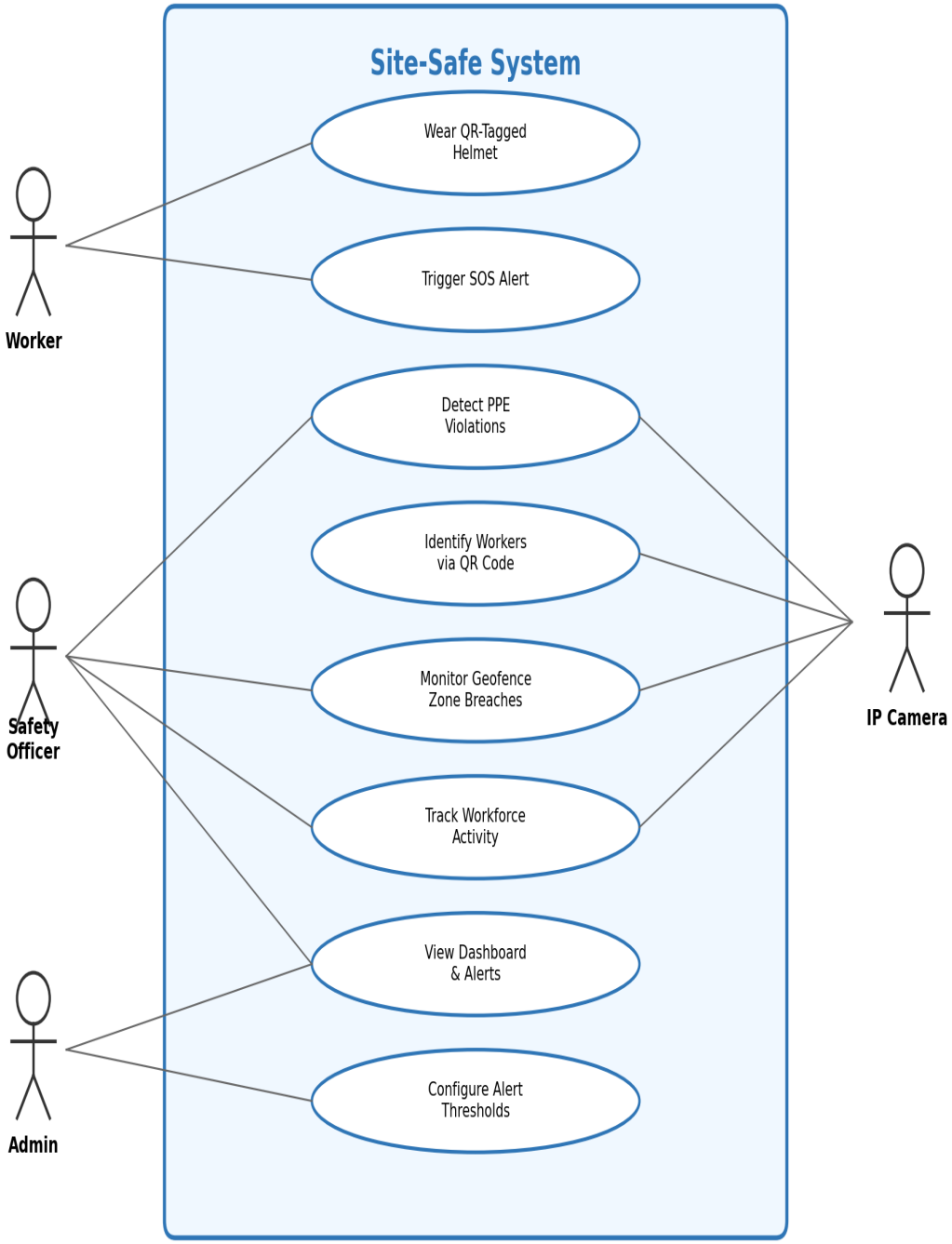



***Figure 3.1:*** *Use Case Diagram*




***Figure 3.2:*** *System Context Diagram*

## 3.2 Architecture

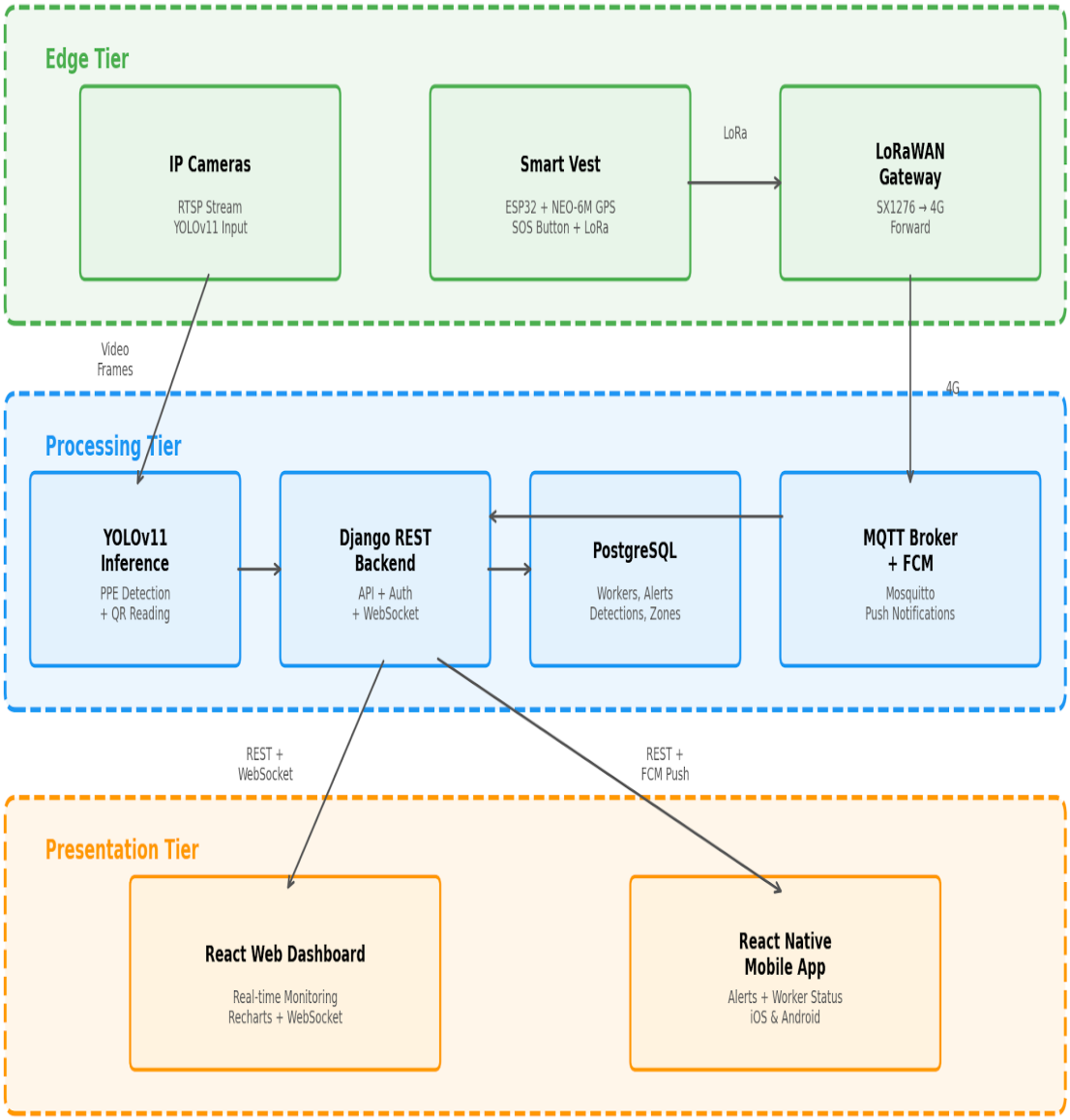



***Figure 3.3:*** *Three-Tier System Architecture*

### 3.2.1 Three-Tier Model

Tier 1 (Edge): IP cameras feed an edge computer that runs a multi-model inference pipeline locally. A Raspberry Pi 5 or Jetson-class unit was proposed; the prototype runs on a laptop GPU (Section 4.1). The primary YOLOv11 model handles PPE detection on every frame; a person-and-vehicle detector and QR code reading run on every third frame, a YOLOv11-

Pose model runs on every third frame for fall detection, and a fire and smoke model runs on every fifth frame. Together these models push violation alerts, fall events, fire detections and worker identification data to the backend, and the same pipeline feeds workforce activity monitoring. A lightweight ESP32-based Smart Vest is proposed to provide supplementary GPS location and SOS alerts via LoRa.

Tier 2 (Network): Camera alerts and QR identification data are sent over HTTP to the backend API, authenticated with per-camera device tokens and buffered in a local outbox while the server is unreachable. For the supplementary Smart Vest, a LoRa receiver is proposed to forward GPS and SOS data via MQTT; an MQTT broker (Mosquitto) handles the publish-subscribe routing, and the vest gateway can also post over HTTP.

Tier 3 (Application): A Django REST backend processes all incoming data, applies business rules (threshold comparisons, geofence polygon checks, activity tracking logic), and persists results to PostgreSQL. Real-time updates are pushed to the React.js web dashboard via Django Channels WebSockets, and critical alerts are forwarded to the React Native mobile app through Firebase Cloud Messaging push notifications.

### 3.2.2 Data Flow

1. Camera captures live video → YOLOv11 on edge device detects PPE violations and reads QR codes per-frame → Violations are tagged with worker identity from QR → JSON alert payloads sent via HTTP POST to backend API → Backend stores event with worker ID, generates compliance record, and pushes to dashboard via WebSocket and to mobile app via FCM.

2. Smart Vest GPS module samples location at configured intervals → Data transmitted via LoRa to gateway → Gateway publishes to MQTT broker → Backend subscriber consumes and processes → SOS button press or location anomaly generates alert → Dashboard and mobile app updated simultaneously. 3. Camera frame captured → YOLOv11-Pose analyses body keypoints (every 3rd frame) → Fall detected if collapsed posture persists 3+ seconds → CRITICAL alert dispatched. 4. Camera frame captured → Fire/smoke model runs (every 5th frame) → Detection checked against zone hot

work permits → Suppressed if permit active and flame not growing, otherwise CRITICAL alert dispatched.

3. Admin interacts with dashboard or mobile app → Uploads floor plans, draws Red Zone polygons, configures alert thresholds → System stores zone geometry and settings → Vision module checks real-time worker positions against active zones.

## 3.3 Module 1: Vision Intelligence

### 3.3.1 PPE Detection

The vision module uses a YOLOv11-M model to continuously inspect camera frames for ten detection classes covering helmets, high-visibility vests, boots, gloves and goggles, each in a worn and a missing state. Frames are captured from IP cameras over RTSP or HTTP streams, resized to 640×640 and passed through the model on the edge device. Detections below a candidate confidence threshold (0.15 in the prototype, pending per-class calibration) are discarded. OpenCV's QRCodeDetector scans sampled frames for QR codes on worker helmets, linking a detected person to their registered identity. A worker without required PPE triggers a violation event logged with worker identity (from QR, when readable), timestamp, camera ID, zone and detection confidence, creating an auditable per-worker compliance record.

### 3.3.2 Geofencing and Intrusion Detection

Administrators define Red Zones by drawing polygon boundaries directly on the site map in the dashboard, and the polygons are stored in the database. Worker positions are checked against active zone polygons with a point-in-polygon test; in the prototype the positions come from vest GPS, because a single camera gives no ground coordinates. An incursion into a restricted zone triggers an immediate alert on the dashboard and a push notification to the safety officer through the mobile app.

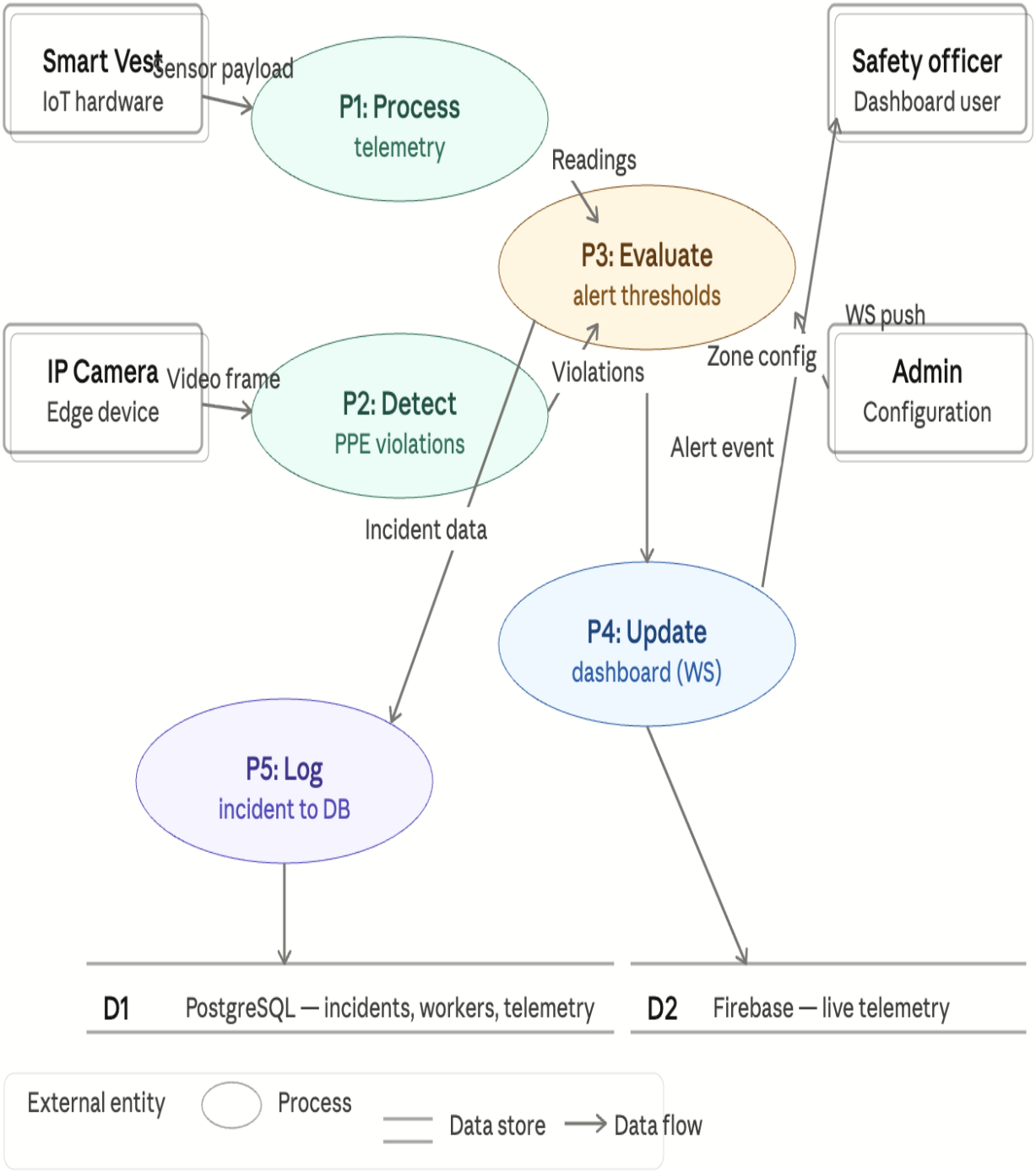



***Figure 3.4:*** *Level-1 Data Flow Diagram*

### 3.3.3 Vehicle Proximity Warning

A parallel detection class identifies heavy construction vehicles: cranes, dump trucks, forklifts, bulldozers. When a vehicle bounding box and a person bounding box appear within a defined proximity threshold, a proximity warning is generated for that camera zone. This catches the blind-spot hazard that accounts for a meaningful proportion of construction vehicle accidents.

### 3.3.4 Workforce Activity Monitoring

Beyond safety compliance, the vision pipeline is extended to serve as a workforce activity monitoring layer. Using the same YOLOv11 person detection output, the system tracks three key metrics. Activity tracking logs when each detected worker enters and exits designated work zones, producing timestamped records of zone presence duration. Break detection identifies workers who remain stationary beyond a configurable threshold (default: 15 minutes) or who leave their assigned zone for extended periods, flagging these as break events. Inactivity monitoring raises an alert when a worker shows no detectable movement for a prolonged duration, which may indicate injury, distress, or a safety concern requiring immediate investigation.

All activity data is stored in PostgreSQL with worker ID, zone ID, and timestamp fields, making it queryable by date range, worker, or zone. This data feeds into both the web dashboard (where it appears as activity charts and zone utilization heat maps) and the mobile app (where supervisors can check any worker's current status and recent activity history).

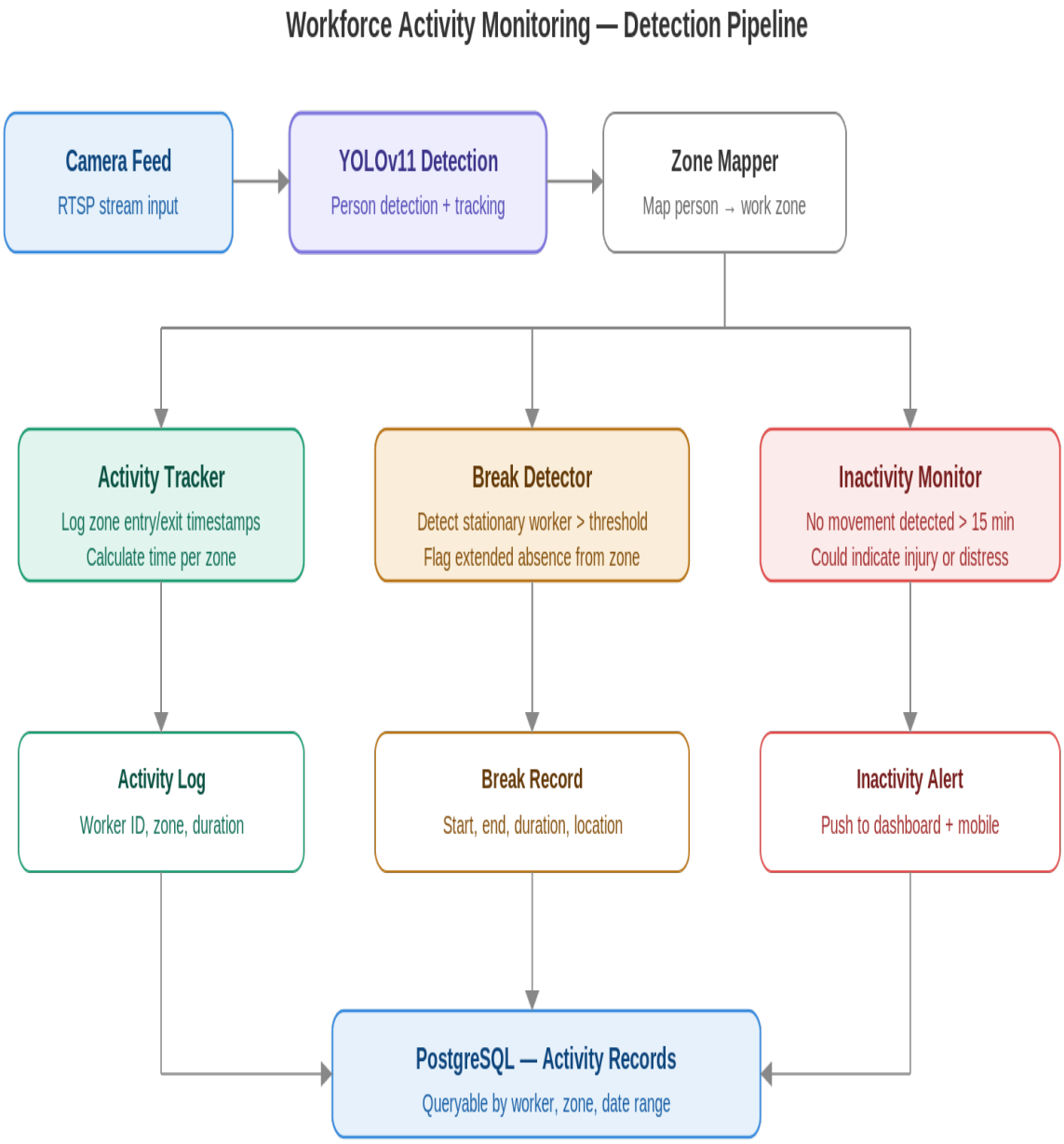



***Figure 3.5:*** *Workforce Activity Monitoring Pipeline*

### 3.3.5 Fall Detection

A second inference model, YOLOv11-Pose (6.4M parameters), runs alongside the PPE detector on every third frame to conserve compute. The pose model outputs 17 body keypoints per detected person: nose, eyes, ears, shoulders, elbows, wrists, hips, knees, and ankles. Fall detection relies on two geometric signals derived from these keypoints. First, the system measures the ratio of a person's horizontal span to their vertical span; a standing

person is taller than wide, while a fallen person is wider than tall. Second, it checks whether the hip keypoints have dropped to or below knee level, indicating the torso has collapsed.

To reduce false positives from workers crouching, bending to pick up materials, or sitting during breaks, a three-second persistence filter is applied. A fall event is only raised if the collapsed posture persists for at least three consecutive seconds. This filters out the brief postural changes that are normal on a construction site. When a genuine fall is confirmed, the system generates a CRITICAL alert with the detection confidence score, duration, and a bounding box snapshot of the frame, dispatched instantly through WebSocket and FCM push notifications.

### 3.3.6 Fire and Smoke Detection

A dedicated fire and smoke detection model runs on every fifth frame from the same IP camera feeds. The model is a YOLO variant fine-tuned on construction-environment fire and smoke datasets, outputting two classes: fire and smoke. Each detection includes a bounding box, confidence score, and estimated region area in pixels.

A key challenge on construction sites is distinguishing between controlled fire activities (welding, cutting, grinding) and genuine uncontrolled fires. Site-Safe addresses this through zone-aware suppression. Each zone in the system can be tagged with permitted activities such as welding or hot work. Additionally, safety officers can issue time-bound hot work permits through the dashboard, specifying a zone, start time, and end time. When the fire model detects flame or smoke in a zone with an active hot work permit, the alert is either suppressed entirely or downgraded to LOW severity and logged without triggering a push notification.

A secondary heuristic tracks whether the detected fire region is growing across consecutive frames. A welding torch produces a small, steady flame that does not expand, while an uncontrolled fire grows over time. If the bounding box area increases by more than 50 percent over a rolling window of 30 frames, the system treats it as a genuine fire regardless of any active hot work permit, escalating the alert to CRITICAL with full push notification dispatch.

## 3.4 Module 2: QR Identification and Supplementary IoT

### 3.4.1 QR-Based Worker Identification

The primary method of identifying individual workers on site is through QR codes printed on their helmets. During registration, the admin assigns each worker a unique QR code generated by the system. The QR is printed as a weatherproof sticker and attached to the top or side of the worker's helmet, visible to overhead and angled cameras. OpenCV's built-in QRCodeDetector runs on each frame alongside YOLOv11, decoding any visible QR and mapping it to the worker's database record. This approach requires no electronics on the worker, costs virtually nothing per unit, and survives the dust, water, and impact conditions of a construction site. The trade-off is that QR readability depends on camera distance and angle, so the system works best at designated checkpoints and work zones where cameras are positioned within 5–10 metres.

***Table 3.1:*** *Worker Identification and Supplementary Sensing*

| Component | Model | Function | Interface | Sample Rate |
|---|---|---|---|---|
| GPS | NEO-6M | Latitude, longitude | UART | 1 Hz |
| Panic Button | Tactile Switch | Manual SOS | GPIO Interrupt | Event-driven |

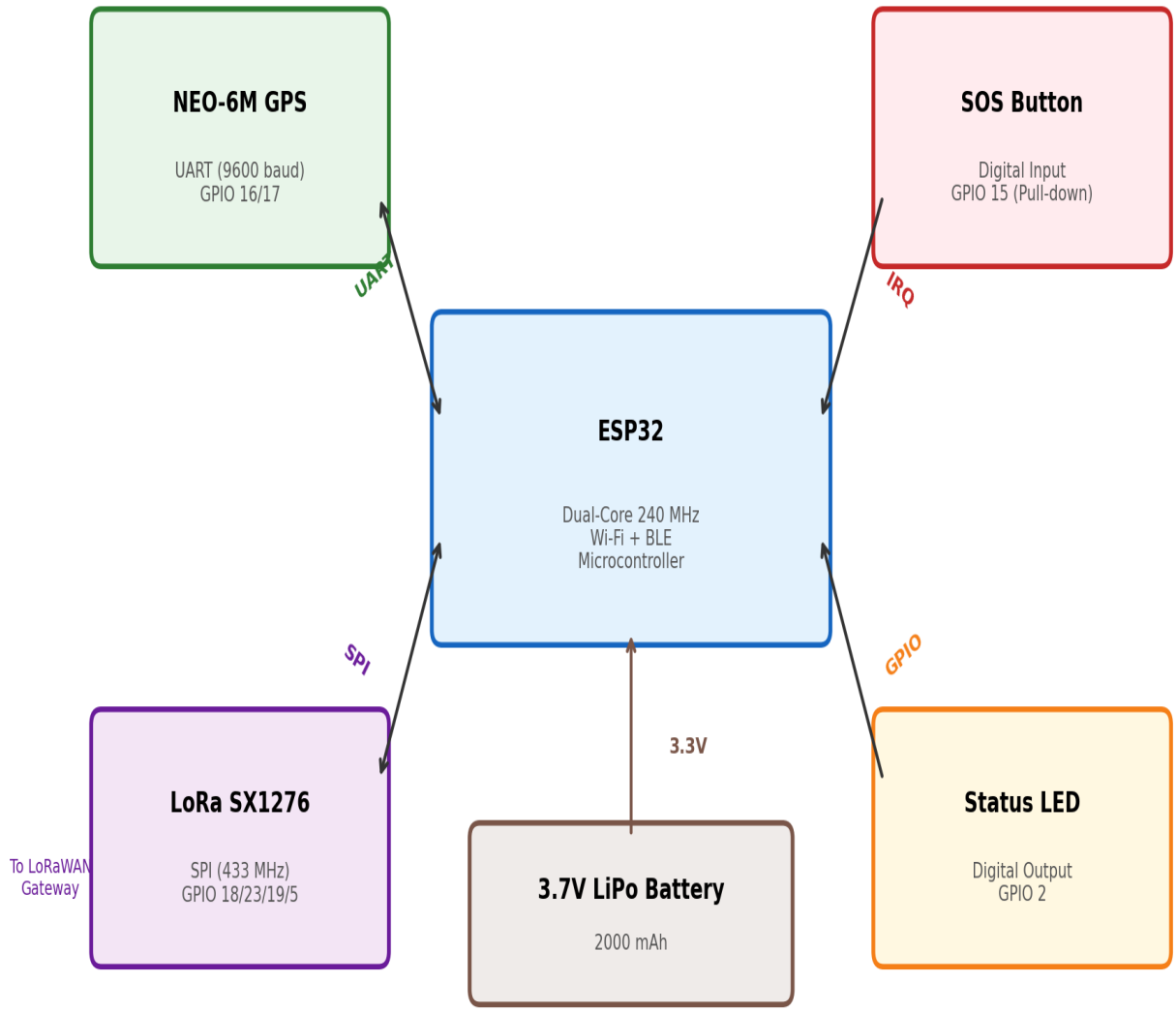



***Figure 3.6:*** *Smart Vest Block Diagram*

Power is supplied by a 3.7V 2500mAh LiPo battery managed through a TP4056 charge controller with overdischarge protection. With only GPS and LoRa active at 30-second transmit intervals, estimated battery life exceeds 24 hours, comfortably covering a full construction shift without the power drain of multi-sensor sampling.

### 3.4.2 Supplementary Smart Vest

For areas outside camera coverage, a lightweight Smart Vest built around the ESP32 microcontroller provides supplementary worker tracking. The vest is deliberately minimal: a GPS module (NEO-6M) for location tracking and a tactile SOS panic button for emergencies.

This design reflects the project's camera-first philosophy—the vest exists to fill gaps, not to serve as the primary monitoring layer. The SOS button triggers an immediate high-priority alert through all channels when pressed, and the GPS module transmits the worker's coordinates at configurable intervals so the system can track location in zones without camera coverage.

By keeping the vest hardware minimal, we avoid the cost, complexity, and battery-life problems that plague multi-sensor wearables in construction environments. A GPS module and a button are simple to manufacture, easy to maintain, and require no worker training beyond "press this if you need help." The MQTT communication layer means the system architecture can accept additional sensors in future iterations without any backend changes.

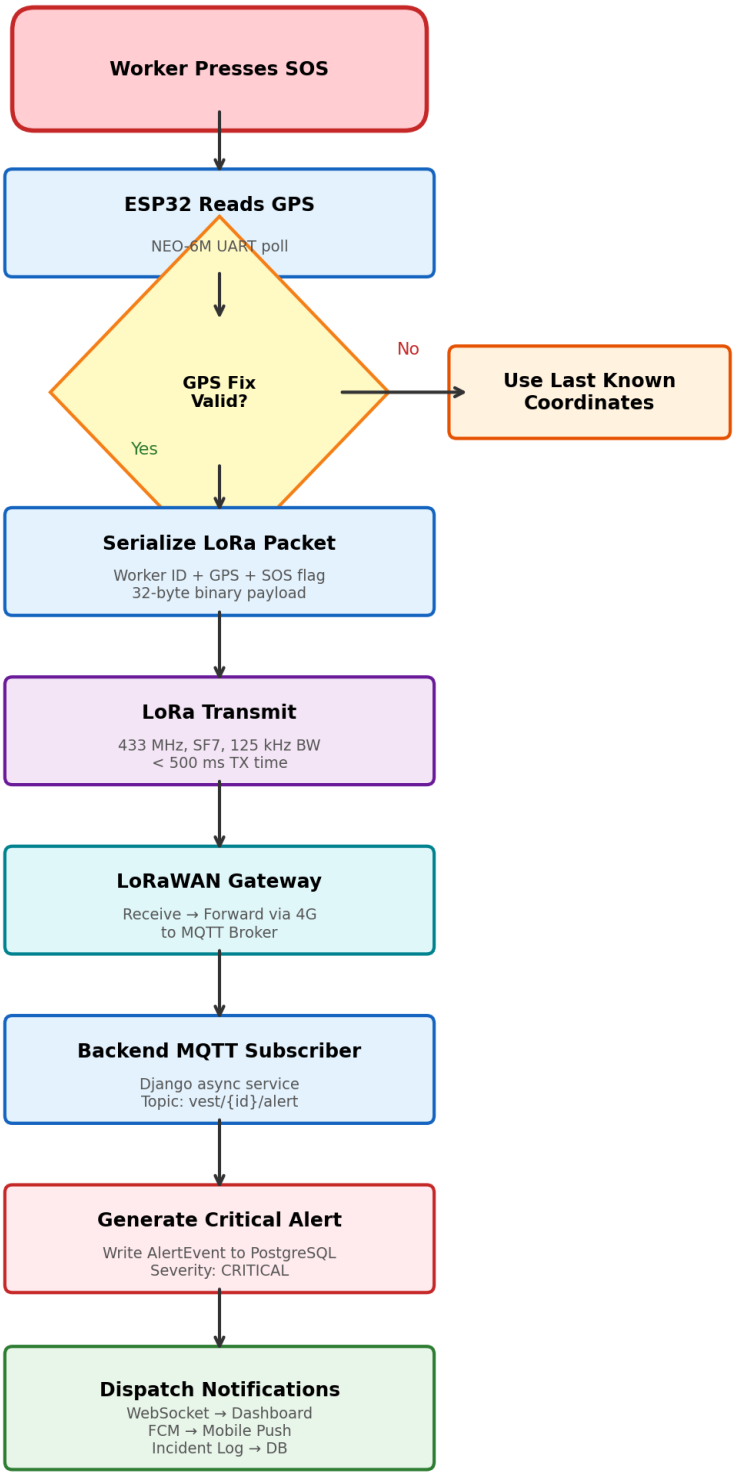



***Figure 3.7:*** *SOS and Location Alert Flowchart*

### 3.4.3 LoRa Communication Layer

The vest transmits GPS coordinates and SOS alerts over LoRa at 433 MHz using an RA-02 transceiver module. LoRa was selected because it achieves 2 to 15 km range in open terrain at very low power consumption, which suits large construction sites where Wi-Fi infrastructure is impractical. Data is serialized into a compact binary packet and transmitted every 30 seconds under normal conditions, with immediate transmission triggered by an SOS button press.

## 3.5 Module 3: Digital Twin Command Center

### 3.5.1 Backend

The Django REST Framework backend provides the API layer that edge devices write to and both the dashboard and mobile app read from. A separate MQTT consumer process handles incoming vest telemetry, and critical alerts are routed to WebSocket clients and to the FCM push notification service. Notifications are sent from a background thread pool after each alert is committed, and PDF and DOCX reports are produced by a separate Node.js report service, so neither blocks API response times.

### 3.5.2 Frontend Dashboard

The React.js dashboard is organized around a central site view with worker avatars at their current GPS-derived positions, color-coded by status: green for normal, yellow for a warning condition, red for an active alert. Surrounding the main view are panel components for the live alert feed, worker status cards, activity monitoring charts, and incident history. All data updates arrive over WebSocket and trigger targeted component re-renders rather than full page refreshes.

## 3.6 Module 4: Mobile Application

### 3.6.1 App Architecture

The mobile application is built with React Native, enabling a single JavaScript codebase to target both iOS and Android platforms. React Native was chosen specifically because the

team already has React expertise from the web dashboard development, and the component model and API service patterns carry over between web and mobile.

The app communicates with the same Django REST backend used by the web dashboard. Authentication uses JWT tokens stored securely in the device keychain. Real-time alert delivery uses Firebase Cloud Messaging (FCM), which provides reliable push notifications even when the app is backgrounded or the device is locked. For live data when the app is in the foreground, a WebSocket connection provides the same real-time stream as the web dashboard.

### 3.6.2 Core Features

The mobile app provides four primary capabilities. First, a live alert feed showing all active alerts with severity color-coding, source identification (camera or vest), and timestamp, with one-tap navigation to detailed alert views. Second, a worker status screen showing all registered workers with their current location zone, compliance status, and status badge (active, warning, SOS, offline). Third, push notifications for critical alerts (PPE violations, SOS, inactivity, zone breaches) that function even when the app is not in the foreground. Fourth, an activity summary view where supervisors can check any worker's zone history, violation record, time worked, and break periods for the current shift.

### 3.6.3 Admin Alert Management

For users with Admin role privileges, the mobile app exposes an alert configuration interface. Admins can enable or disable specific alert categories (PPE violations, fall detection, fire and smoke, SOS alerts, zone breaches, inactivity alerts), set custom threshold values for each category (inactivity timeout, zone boundary parameters, fall persistence duration), manage hot work permits for zones where welding or cutting is scheduled, configure notification routing (which roles receive which alert types), and set shift schedules that determine when monitoring is active. All configuration changes sync immediately to the backend and take effect across the entire system.

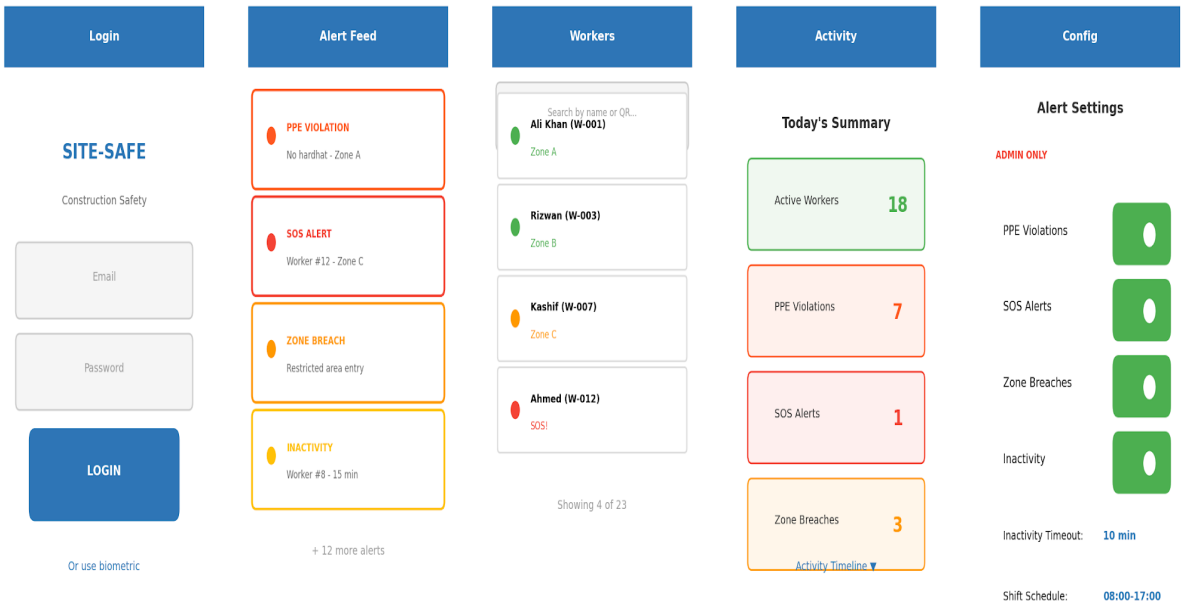



Figure 3.8: Mobile Application Screen Wireframes

***Table 3.2:*** *Mobile App Feature Matrix*

| **Feature** | **Safety Officer** | **Admin** | **Description** |
|---|---|---|---|
| Live alert feed | Yes | Yes | Real-time list of all active alerts with severity and source |
| Worker status view | Yes | Yes | Current location, compliance status, and alerts for all workers |
| Push notifications | Yes | Yes | FCM alerts for SOS, PPE violations, zone breaches |
| Activity summary | Yes | Yes | Zone history, time worked, break periods per worker |
| Enable/disable alerts | No | Yes | Toggle specific alert categories on/off |
| Set thresholds | No | Yes | Configure inactivity timeout, zone boundaries, alert routing |
| Notification routing | No | Yes | Assign alert types to specific roles |
| Shift scheduling | No | Yes | Define active monitoring windows |

## 3.7 Admin Module

The admin interface, accessible through both the web dashboard and the mobile app, handles site configuration and worker management. Admins can register workers and assign QR codes to their profiles, draw geofence zones on the site map, configure alert and notification preferences, and generate or download PDF compliance reports for any date range. Access is role-gated: Admin, Safety Officer, and read-only Viewer roles are defined with corresponding permission sets enforced at the API level.

## 3.8 Limitations

QR code identification depends on camera distance and angle: at ranges beyond approximately 10 metres, or when helmets are tilted away from the camera, the QR may not be readable. Strategic camera placement at key checkpoints mitigates this, but full-site individual tracking is not guaranteed. Continuous camera monitoring raises privacy concerns that require transparent worker consent processes and clear data handling policies. The supplementary Smart Vest provides GPS and SOS coverage outside camera zones, but relies on LoRa signal propagation which can be degraded by dense metal scaffolding. The workforce activity monitoring component can only track workers in areas with camera coverage.

# Chapter 4

# Hardware and Software Implementation

## 4.1 Development Environment

Site-Safe was developed and measured on one Apple M4 laptop with 16 GB of memory running macOS; the versions in Table 4.1 were read from the project environments and lock files on 20 October 2026. The PPE model was trained on a Google Colab Tesla T4 GPU, because training on the laptop ran out of memory. A docker-compose file and a GitHub Actions workflow are written but have not yet been run.

Table 4.1: Development Environment Summary

| Tool / Platform | Version | Purpose |
|---|---|---|
| Python | 3.14.3 | Backend, camera pipeline and measurement tools |
| Django / Django REST Framework | 6.0.4 / 3.17.1 | REST API, permissions and serializers |
| Django Channels / Daphne | 4.3.2 / 4.2.3 | WebSocket delivery; ASGI server for REST and WebSocket |
| PostgreSQL | 18.3 | Development and measurement database; backend tests also run on SQLite |
| Redis | 7 (container image) | Channel layer between backend processes in the compose deployment |
| Eclipse Mosquitto / paho-mqtt | 2 (container image) / 1.6.1 | Vest gateway messages read by the MQTT consumer |
| Ultralytics / PyTorch / OpenCV | 8.4.41 / 2.11.0 / 4.13.0 | Detection, pose estimation and QR decoding |
| Node.js / Express | 25.8.1 / 5.2.1 | Report service runtime |
| docx / pdfkit | 9.7.1 / 0.20.2 | DOCX and PDF report rendering |
| React / Vite / Tailwind CSS | 19.2.8 / 8.2.2 / 4.3.3 | Web command centre |
| Leaflet / react-leaflet | 1.9.4 / 5.0.0 | Site map and zone editor (OpenStreetMap tiles) |
| Expo SDK / React Native | 58.0.12 / 0.88.0 | Mobile application |
| Docker Compose / GitHub Actions | Written, not yet run | Eight-service deployment; CI running every test suite |
| Git | 2.53.0 | Version control |

## 4.2 QR Identification and Smart Vest

### 4.2.1 QR Worker Identification

Each worker has a random badge code, created at registration and reissued from the Workers page, which prints a QR badge with the payload SSW1:<code> for the helmet. On the camera, identity.py decodes QR codes with OpenCV on the frames sampled for the person detector and associates each badge with the person box that contains it, following that person between samples by box overlap. A badge inside two person boxes, or two

badges inside one box, is treated as unknown rather than guessed, so people standing close together are not misattributed.

Alerts from a person with a known badge are attributed to that worker; others are marked as not identified. The same badges feed attendance (Section 4.7), with at most one sighting per badge per minute, and unknown or revoked codes are ignored with HTTP 202 so that they cannot block a camera's queue. Thirteen camera-side tests, including decoding of real OpenCV-encoded images, and seven backend tests cover decoding, association and attribution. Reading range, angle and lighting have not yet been measured with printed badges, so the 5–10 metre range in Section 3.4.1 remains an assumption.

### 4.2.2 Smart Vest

The physical Smart Vest (ESP32 [13], GPS module, SOS button and LoRa link [14]) has not been built. It was moved to future work through a scope-change request to the supervisor: the vest is supplementary by design (Section 3.4.2), parts and lab access were not available in the time left, and the camera functions carry the primary objectives. Figure 3.6 and Tables 4.2 and 4.3 remain the proposed hardware design.

The software path a vest would use is implemented. A gateway posts JSON telemetry (vest, position, battery and SOS state) to /api/v1/vests/telemetry/ with a gateway token, or publishes it to sitesafe/vest/<id>/telemetry and /sos on a Mosquitto broker [15] read by the MQTT consumer. Both paths validate the payload and share one handler (Figure 4.8). A vest simulator drives this path with scripted scenarios; its token is flagged, so its alerts are labelled SIMULATED, create no attendance and are excluded from reports.

Table 4.2: Proposed Smart Vest Component Specifications

| Component | Model | Range | Accuracy | Interface |
|---|---|---|---|---|
| GPS | NEO-6M | Worldwide | 2.5 m CEP | UART |
| SOS Button | Tactile Switch | N/A | N/A | GPIO |

Table 4.3: Proposed ESP32 Pin Allocation

| GPIO Pin | Connected To | Protocol | Notes |
|---|---|---|---|
| GPIO 16 (RX2) | NEO-6M TX | UART | 9600 baud |
| GPIO 17 (TX2) | NEO-6M RX | UART | 9600 baud |
| GPIO 18 | LoRa Module SCK | SPI | SX1276 |
| GPIO 23 | LoRa Module MOSI | SPI | SX1276 |

| GPIO Pin | Connected To | Protocol | Notes |
|---|---|---|---|
| GPIO 19 | LoRa Module MISO | SPI | SX1276 |
| GPIO 5 | LoRa Module NSS | SPI | Chip select |
| GPIO 2 | Status LED | Digital | Activity indicator |
| GPIO 15 | SOS Button | Digital | Pull-down, interrupt |

## 4.3 PPE Detection Module

The camera pipeline runs the PPE detector models/sitesafe_v5.pt, a YOLO11-medium model trained with Ultralytics [3], on every frame. Because the PPE model has no person or vehicle class, a COCO-pretrained YOLO11-nano detector supplies person and vehicle boxes on every third frame, and a YOLO11-nano pose model runs on every third frame for fall detection. All models read the same unannotated frame; boxes are drawn only on a display copy. Each missing-PPE box is tracked as one person's episode (ppe_episodes.py). Same-class boxes covering one body part are first merged (overlap of 0.4 IoU, or one centre inside the other at 0.2), because the detector can report one body part two or three times. An episode alerts once after persisting for one second in at least three frames, and again only after it has been unseen for 30 seconds, so a flicker raises nothing and two people missing helmets raise two alerts.

### 4.3.1 Dataset and Classes

The training data is the Roboflow dataset ppe-detection-qlq3d (CC BY 4.0) [4]: 5,140 images in ten classes, helmet, vest, boots, gloves and goggles, each with a matching no-class, so that a violation is detected directly rather than inferred. It has no person or vehicle class, hence the second detector.

The published split (Table 4.4) is not independent. The repository audit (audit_dataset.py) found 573 source groups and 30 groups of exact duplicates with members in more than one split, and 69 groups of identical images with conflicting labels. Test images therefore have near-copies in training, which inflates the scores below. A group-constrained re-split (3,598 / 771 / 771 images) has been generated but is held back until the label conflicts are resolved.

Table 4.4: Dataset Split Configuration

| Split | Images | Percentage | Purpose |
|---|---|---|---|
| Training | 3,597 | 70% | Model weight optimization |
| Validation | 1,026 | 20% | Early stopping and model selection |

| Split | Images | Percentage | Purpose |
|---|---|---|---|
| Test | 517 | 10% | Final performance evaluation |

### 4.3.2 Training Configuration

The model was fine-tuned from COCO-pretrained YOLO11-medium weights with the settings in Table 4.5, recorded in the exported run manifest, replacing an earlier nano model trained at 416 pixels (mAP@0.5 0.544). The team's Colab record shows early stopping after 102 of 150 epochs, with the best weights at epoch 77. Figure 4.1, drawn from the exported run, summarises the path from data to deployed weights.

Table 4.5: YOLOv11 Training Hyperparameters

| Parameter | Value | Rationale |
|---|---|---|
| Base model | YOLO11-m (medium), COCO-pretrained | Larger capacity than the earlier nano baseline |
| Input resolution | 640 × 640 | Standard Ultralytics training resolution |
| Batch size | 16 | Fits the T4's memory at 640 px |
| Epochs | 150 budget, early stopping (patience 25) | Recorded run stopped after 102 epochs; best at epoch 77 |
| Optimizer | AdamW | Weight decay handling |
| Learning rate | 0.001, linear decay to 1% (lrf 0.01) | Cosine schedule disabled for this run |
| Augmentation | Mosaic, flip, HSV, scale (defaults) | Mosaic disabled for the final 10 epochs |
| Seed | 42, deterministic | Repeatable run |
| Confidence threshold | 0.15 | Runtime candidate threshold; not yet calibrated per class |
| NMS IoU threshold | 0.7 | Ultralytics default |

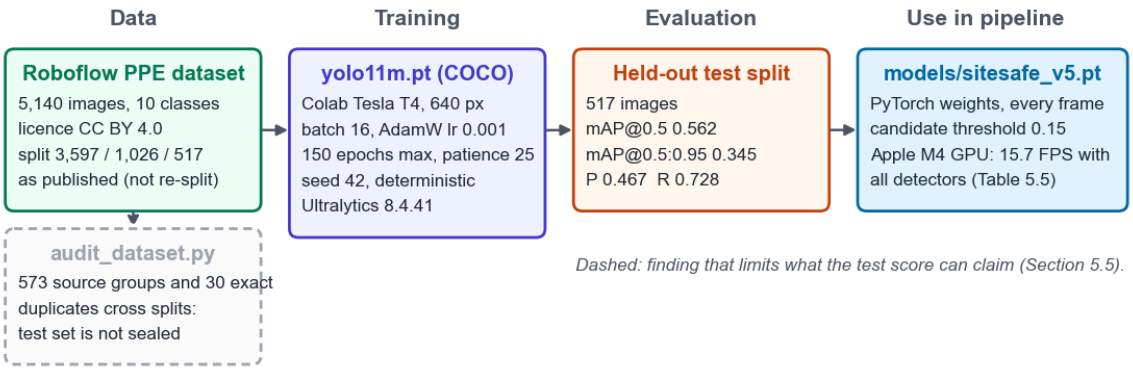



Figure 4.1: PPE Model Training and Evaluation Pipeline

### 4.3.3 Model Performance

On the held-out split the model reaches mAP@0.5 of 0.562 and mAP@0.5:0.95 of 0.345, with precision 0.467 and recall 0.728 (Table 4.6). Equipment that is worn is recognised reliably: helmet, vest and boots exceed 0.95. Missing equipment, the condition the system exists to detect, is recognised poorly: no-gloves 0.062, no-boots 0.130 and no-helmet 0.309. Absence must be inferred from where equipment would be, and the negative classes are under-represented. The larger model did not fix this, so better data for the weakest classes is the most valuable improvement, and because of the split leakage every figure is an upper bound.

Table 4.6: Per-Class Detection Performance

| Class | Precision | Recall | mAP@0.5 |
|---|---|---|---|
| helmet | 0.893 | 0.959 | 0.958 |
| no-helmet | 0.249 | 0.714 | 0.309 |
| vest | 0.853 | 0.954 | 0.962 |
| no-vest | 0.578 | 0.879 | 0.782 |
| boots | 0.845 | 0.998 | 0.977 |
| no-boots | 0.127 | 0.596 | 0.130 |
| gloves | 0.222 | 0.316 | 0.276 |
| no-gloves | 0.109 | 0.323 | 0.062 |
| goggles | 0.470 | 0.793 | 0.635 |
| no-goggles | 0.321 | 0.750 | 0.530 |
| all classes | 0.467 | 0.728 | 0.562 |

### 4.3.4 Fall Detection

Fall detection (fall_detection.py) runs its own pose model on every third frame, since neither the PPE nor the COCO detector gives keypoints. A posture counts as collapsed only when both signals of Section 3.3.5 agree: the confident keypoints form a box wider than tall, and the hips are at or below the knees. If either signal cannot be measured, for example with a leg out of frame, the result is not-collapsed, so missing data does not raise an alarm.

People are tracked between samples by a confidence-weighted keypoint centroid, and a fall is confirmed only after three continuous seconds of collapse. It raises one FALL alert and latches until the posture clears; brief missed detections are tolerated. The alert names a worker only when a QR badge is associated with the track. Twenty-four tests cover the geometry and the tracker, and on real pose output from photographs, standing pedestrians were read as not collapsed. The module has not been tested on fall footage, and its pixel geometry is uncalibrated per camera.

### 4.3.5 Fire and Smoke Detection with Hot-Work Permits

On the camera, fire_detection.py runs on every fifth frame and confirms fire or smoke only when it persists for two seconds with gaps of at most one second, ignoring specks below 0.05 percent of the frame; after an alert it latches until the scene has been clear for 30 seconds, so a fire raises one alert, not one per frame.

Severity is set by the server (alerts/fire_policy.py) from hot-work permits, which safety officers issue for a zone and time window on the Zones page and can revoke. Without an active permit FIRE is CRITICAL and SMOKE is HIGH; within one, FIRE is MEDIUM and SMOKE LOW, and the alert is linked to the permit. Unlike the suppression proposed in Section 3.3.6, a permitted flame is still stored and broadcast, so a welding fire that spreads is not hidden, and FIRE cannot be disabled. The growth heuristic was not implemented, as no labelled footage exists to set it.

The fire model itself is not yet trained: a preparation script and a Colab notebook for a two-class YOLO11-small detector exist, but the dataset and licence must be confirmed first, and the pipeline runs without the stage until the weights exist. The permit policy was verified end to end on 20 October 2026 with injected events: a fire in a permitted zone became MEDIUM and was linked, the same fire elsewhere CRITICAL.

### 4.3.6 Camera Sources and Reliable Event Delivery

The camera source is a device index or a stream URL, so an RTSP camera, an NVR or a phone running an IP-camera application uses the same code path as a webcam, and up to 30 consecutive failed reads are tolerated before a stream counts as disconnected. Each pipeline has a camera identifier, and zones list their cameras, so each alert is attributed to a zone.

Every event is first written to a local SQLite outbox with a unique event identifier and delivered in the background, retrying with exponential back-off capped at 30 seconds, also after a restart. The server stores each identifier once, answering 201 for a new event, 200 for a duplicate and 202 when the alert type is disabled; a validation error moves the event to dead letters. Cameras use device tokens scoped to their identifier, and read their alert rules every 60 seconds, so a new inactivity timeout set on the dashboard applies without a restart.

### 4.3.7 Vehicle Proximity Detection

Proximity uses the COCO detector's person and vehicle boxes. Without depth, the shortest gap between a person box and a vehicle box is divided by the person's box height, making the threshold roughly scale-invariant; one person-height is about 1.7 metres when both are at a similar distance from the camera. A breach must persist for three evaluated frames, latches until it clears, and is CRITICAL when the boxes overlap and HIGH otherwise.

Nine unit tests cover the geometry, scale invariance and persistence, and with the real detector a still image of people beside a bus produced a CRITICAL alert shown on the dashboard. COCO has no forklift class, so plant is reported as a truck or car, and a worker standing in front of a distant vehicle is over-reported. Live site footage has not been evaluated.

## 4.4 Backend Implementation

The backend is a Django 6 and Django REST Framework [10] application on PostgreSQL 18, served by Daphne so that REST and WebSocket traffic share one ASGI process through Django Channels [22]. Its five applications are accounts, alerts, workers, sitemap (zones and permits) and reports. Table 4.7 lists the endpoints, Table 4.8 the schema and Figure 4.2 the data model, drawn from the Django models.

Users receive a 30-minute JWT access token and a refresh token, which the clients renew silently, and Admin, Safety Officer and Viewer permissions are enforced on every endpoint. Producers use device tokens instead: only a hash of each key is stored, each has a kind (camera or vest gateway) and permitted cameras, and none can sign in. No write endpoint is anonymous and no credentials remain in the code. The WebSocket requires a valid user token and an allowed origin and closes with code 4401 otherwise, which tells clients to renew and reconnect.

Alert, resolution and vest changes are broadcast after the transaction commits, so clients never see a rolled-back alert, and a Redis channel layer carries them between processes. Alert rules set, per type and optionally per zone, whether an alert is enabled, its threshold and whether it notifies; SOS and FIRE cannot be disabled. Notifications go out after commit through Firebase Cloud Messaging [12] and e-mail, respecting user preferences. A presence monitor runs every 30 seconds, marking vests offline after 120 seconds of silence and closing attendance sessions.

Table 4.7: Core API Endpoints

| Endpoint | Method | Description |
|---|---|---|
| /api/token/, /api/token/refresh/ | POST | Sign in; renew the access token |
| /api/v1/profile/, /api/v1/profile/change-password/ | GET/PATCH, POST | Signed-in user's profile and password |
| /api/dashboard/stats/ | GET | Counts and charts from stored records |
| /api/alerts/ | GET/POST | List or create alerts (date range, type, status filters) |
| /api/alerts/{id}/, /api/alerts/{id}/resolve/ | GET/PATCH, POST | Read, update or resolve one alert |
| /api/alerts/{id}/snapshot/ | GET | Evidence frame of a camera alert; signed-in users only |
| /api/alerts/simulate/ | POST | Create a simulated alert (labelled) |
| /api/v1/camera-events/ | POST | Camera event with event_id; device token |
| /api/v1/camera-activity/ | POST | Badge sightings and idle start/end; device token |
| /api/v1/camera-config/ | GET | Alert rules for this camera; device token |
| /api/alerts/ingest/ | POST | Legacy edge ingest; device token |
| /api/workers/, /api/workers/{id}/ | GET/POST, GET/PUT/PATCH/DELETE | Worker registry and vest assignment |
| /api/workers/{id}/badge/ | GET/POST | Read or reissue a worker's QR badge |
| /api/v1/vests/, /api/v1/vests/{id}/ | GET/POST, GET/PUT/PATCH/DELETE | Smart Vest records |
| /api/v1/vests/telemetry/ | POST | GPS, battery and SOS telemetry; gateway token |
| /api/v1/activity/ (+ /timeline/, /daily/, /summary/) | GET | Activity records, per-worker timelines and daily totals |

| Endpoint | Method | Description |
|---|---|---|
| /api/v1/detections/ | GET/POST | Detection records |
| /api/v1/alert-config/ | GET/POST/PUT/PATCH/DELETE | Alert rules: enabled, threshold, notify |
| /api/v1/zones/ (+ /{id}/workers/, /{id}/check_point/) | GET/POST/PUT/PATCH/DELETE | Zones, assigned workers, point-in-zone test |
| /api/v1/hot-work-permits/ (+ /{id}/revoke/) | GET/POST/PUT/PATCH/DELETE | Hot-work permits and revocation |
| /api/v1/fcm-tokens/, /api/v1/fcm-tokens/{token}/ | POST, DELETE | Register or remove a phone's push token |
| /api/v1/reports/ (+ /generate/, /{id}/download/) | GET, POST, GET | List, generate (DOCX or PDF) and download reports |
| /ws/dashboard/?token=<JWT> | WebSocket | Live alert, vest and worker events |

Table 4.8: Database Schema Summary

| Table | Key fields | Relations / notes |
|---|---|---|
| users | id, email, full_name, role, is_active | Roles: Admin, Safety Officer, Viewer |
| service_tokens | id, name, kind, key_hash, camera_ids, is_simulator, is_active | Device tokens; only a hash of the key is stored |
| workers | id, name, role, zone_id, badge_code, last_seen_at, is_active | FK zones; badge_code is unique (QR payload) |
| smart_vests | id, vest_id, worker_id, battery_level, latitude, longitude, sos_active, last_seen | One-to-one with workers; battery unknown until reported |
| zones | id, name, risk_level, boundaries, camera_ids, max_occupancy | Polygon boundaries and camera assignment (JSON) |
| hot_work_permits | id, zone_id, description, starts_at, ends_at, issued_by_id, revoked_at | FK zones, users |
| alerts | id, alert_type, severity, source, worker_id, camera_id, hot_work_permit_id, confidence, model_version, snapshot, event_id, is_resolved | FK workers, permits; event_id unique for idempotent ingest |
| alert_config | id, zone_id, alert_type, is_enabled, threshold_seconds, notify_push, notify_email | One rule per type and zone; SOS and FIRE always enabled |
| activity_logs | id, worker_id, zone_id, action, timestamp, metadata | Check-in/out, idle and zone events at event time |
| detections | id, camera_id, worker_id, class_name, confidence, bbox | FK workers |
| reports | id, report_type, file_format, date_from, date_to, created_by_id | FK users; stored before download |
| fcm_tokens, user_preferences | user_id, token, device_name / notify_push, notify_email | FK users; push targets and opt-outs |

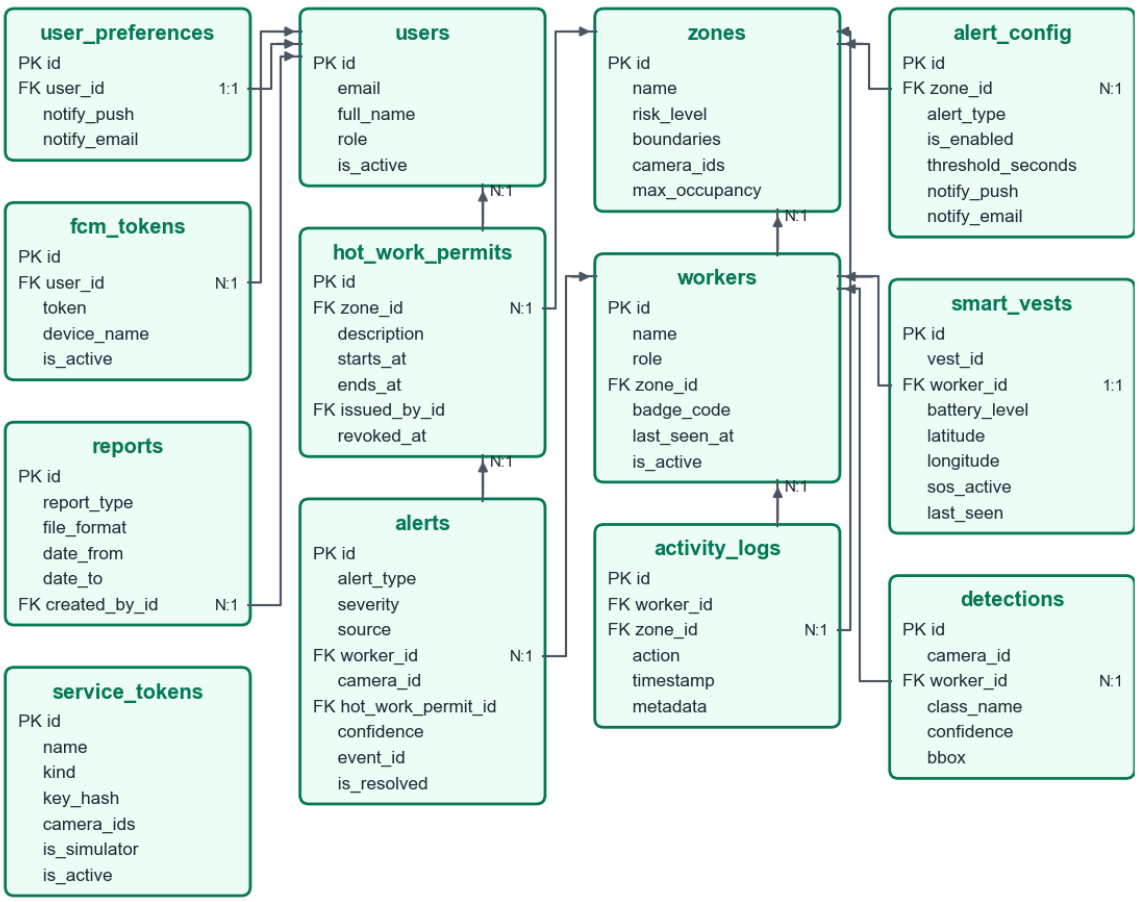



Figure 4.2: Entity Relationship Diagram

## 4.5 Frontend Dashboard Implementation

The web command centre is a React 19 application built with Vite and Tailwind CSS, with routes for the dashboard, workers, incidents, vests, zones, reports and settings. Every dashboard figure comes from stored records; the seeded charts, vitals and compliance percentage of the earlier prototype were removed. A WebSocket adds and removes alerts live, and the client resynchronises after a reconnect.

The Zones page (Figure 4.3, left) shows a Leaflet [23] map on which administrators draw zone polygons, set risk level, occupancy limit and cameras, and manage hot-work permits; vest positions appear coloured by status. The Workers page registers and edits workers, assigns vests, prints badges and opens each worker's daily timeline (Figure 4.3, right). Controls appear only for roles that may use them. Reports are generated in DOCX or PDF for a period in the user's time zone and give counts, resolution times, violation-free days

and attendance, each defined, but no compliance percentage, because the system records violations and not compliant observations, so no denominator exists.

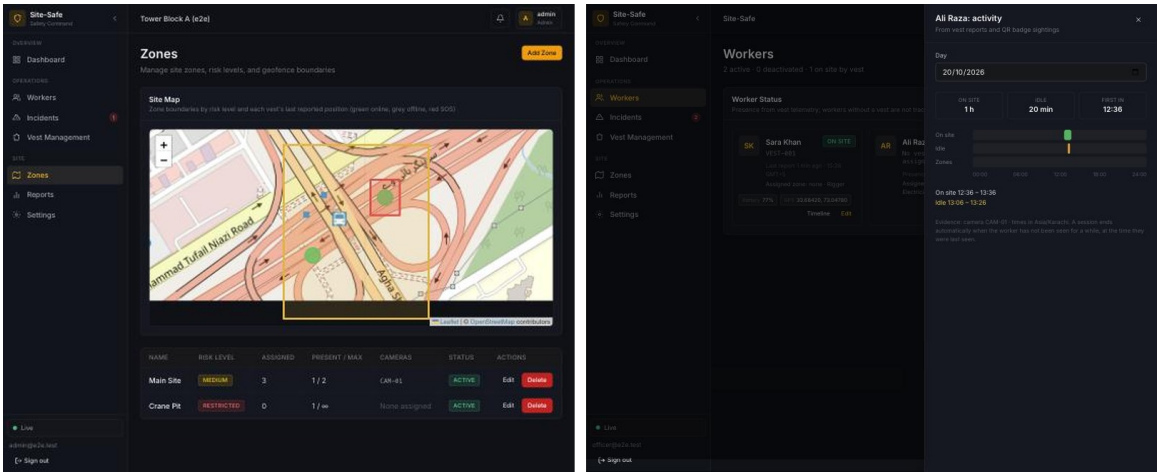



Figure 4.3: Web Command Centre: Zone Map and Worker Timeline

## 4.6 Mobile Application

The mobile application uses React Native 0.88 [11] on Expo SDK 58 in TypeScript, following Section 3.6. Expo builds Android and iOS clients without a local Xcode or Android Studio installation and supplies the FCM device token the backend supports. The app uses the same REST API and WebSocket as the web dashboard.

### 4.6.1 Technology Stack

Table 4.9: Mobile Application Technology Stack

| Component | Technology | Purpose |
|---|---|---|
| Framework | React Native 0.88 on Expo SDK 58 [21], TypeScript | Android and iOS from one codebase |
| Navigation | Expo Router (file-based tabs and stack) | Alerts, Workers and Settings tabs; alert detail |
| State | React context and reducers | Live alert and worker state |
| HTTP client | fetch wrapper with single-flight token refresh | REST calls; sign-out when renewal fails |
| Live updates | WebSocket with 4401 renewal, back-off and resync | Same live stream as the web dashboard |
| Push notifications | expo-notifications 58.1 (FCM on Android) | Alert delivery; tap opens the alert |
| Secure storage | expo-secure-store 58.0 | Tokens in the device keychain/keystore |
| Quality checks | jest-expo (16 tests), ESLint, TypeScript | Run in CI |

### 4.6.2 Core Screens

After sign-in with the server address and credentials, the Alerts tab lists open or all alerts and updates live; the detail screen shows the evidence frame, confidence and model version, and lets Safety Officers and Admins resolve the alert, which other sessions see immediately. The Workers tab shows role, zone, vest and presence, and Settings signs out (Figure 4.4). Reports, zone editing and worker registration stay on the web by design. Of the screens in Figure 3.8 and Table 3.2, the activity summary, rule configuration and shift scheduling were not built.

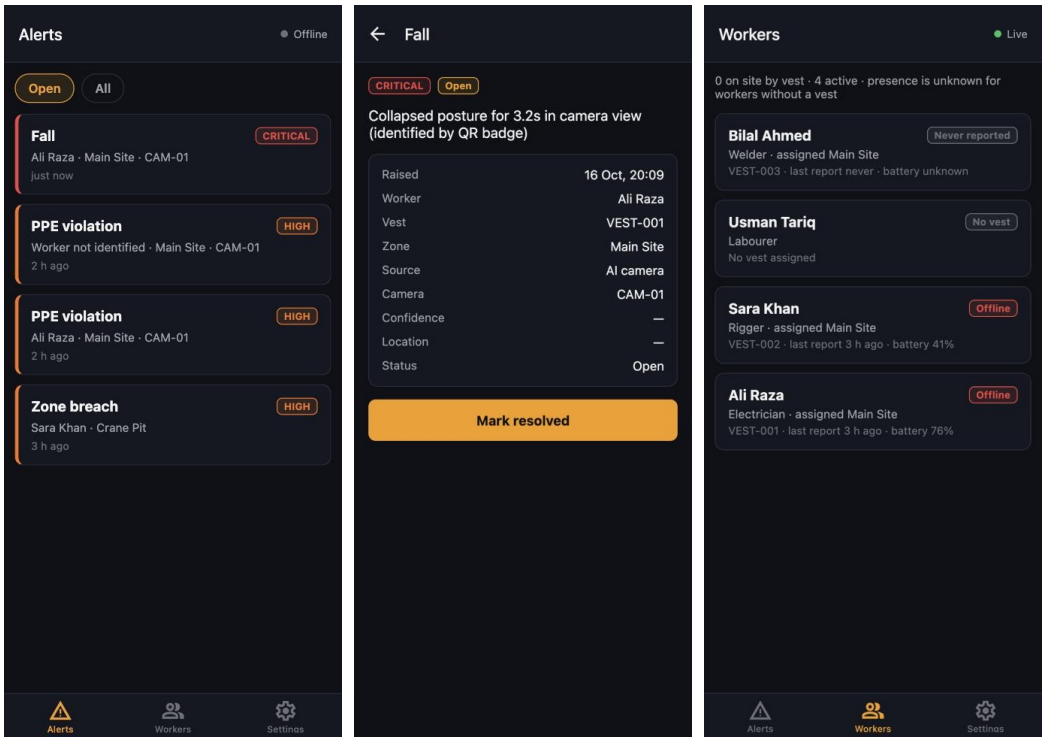



Figure 4.4: Mobile Application: Alerts, Alert Detail and Workers

### 4.6.3 Push Notification Flow

On Android the app registers its FCM token after sign-in and removes it on sign-out; alerts arrive on a high-importance channel and a tap opens the alert. The backend side is tested with a mocked FCM service and the registration logic with unit tests, but delivery to a real phone needs a Firebase configuration and an Android development build and has not been verified. iOS gets no remote push and relies on the live socket. Figure 4.4 was captured from the app's web build, on which sign-in, live alerts, resolution, cross-session updates and sign-out were exercised end to end.

## 4.7 Workforce Activity Monitoring

Inactivity tracking (inactivity.py) follows people between samples by centroid; a track that stays within a movement threshold for the timeout set in the dashboard rules raises one INACTIVITY alert and re-arms only after moving, and brief occlusions do not drop it. Displacement is in uncalibrated pixels.

Identified workers get attendance automatically (workers/activity.py). The first evidence of a worker, a badge sighting or a vest report, checks them in; idle start and end come from the inactivity tracker; and after 30 minutes without evidence the presence monitor checks them out at the time they were last seen. Late events, such as those replayed from an outbox, are merged into their session. A timeline API gives per-worker daily intervals of presence and idleness, shown on the Workers page and in the report attendance tables.

Twenty-three tests cover check-in, check-out, idleness and late events, and an end-to-end run on 20 October 2026 gave 60 minutes on site and 20 idle with check-out at the last sighting, identical in the timeline and the PDF. Records exist only when a worker is seen with a readable badge or wears a vest, and identities can swap when identified people cross; no annotated real-camera scenario has been compared yet.

## 4.8 System Architecture and Deployment

Figure 4.5 shows the implemented components. The camera pipeline and the vest gateway or simulator write through authenticated HTTP or MQTT, the API is the only writer to PostgreSQL, and clients read through REST and receive changes over the WebSocket. The three tiers of Figure 3.3 remain, but notifications use an after-commit thread pool instead of Celery, documents come from a Node.js report service, and the edge runs on a laptop GPU instead of a Raspberry Pi or Jetson.

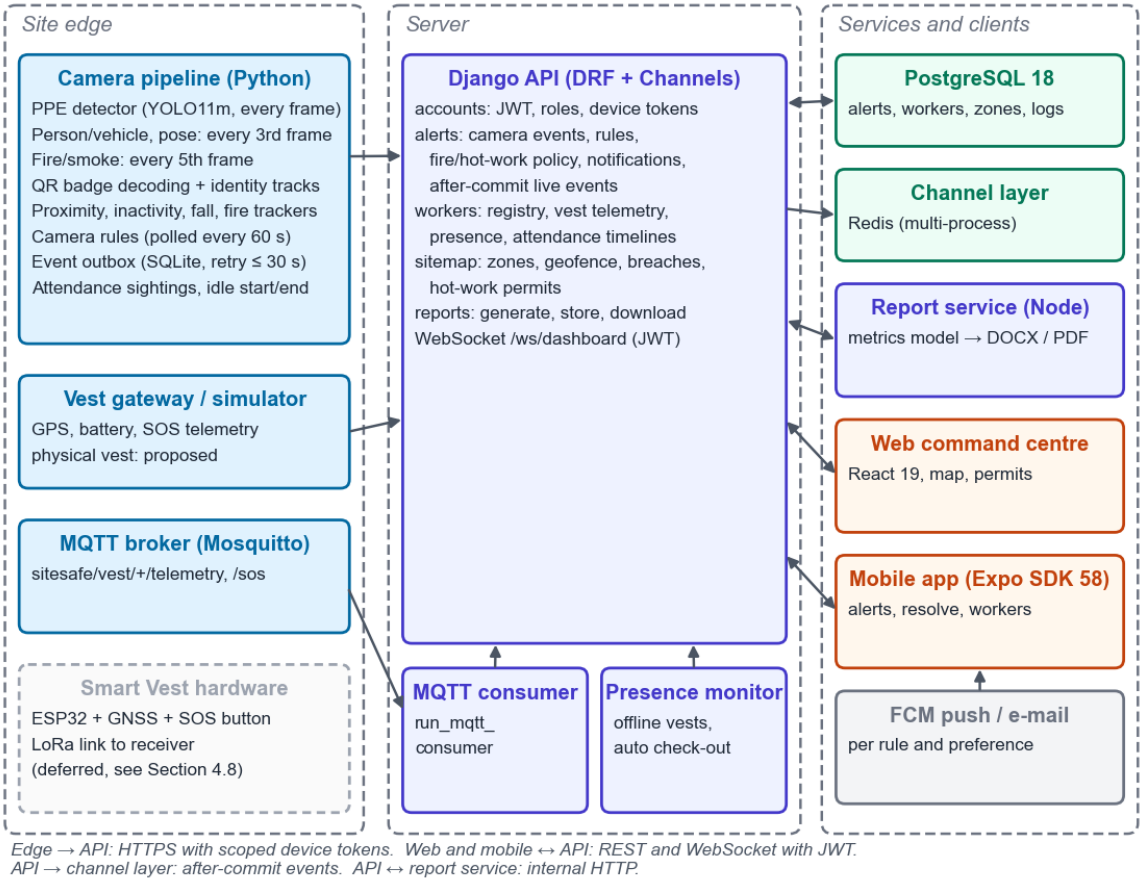



Figure 4.5: Component View of the Implemented System

Figure 4.6 shows the demonstration deployment: the camera pipeline runs natively to use the GPU, and the server is defined as eight docker-compose services. The compose file has not yet been run, so Chapter 5's results come from the same services started on the host.

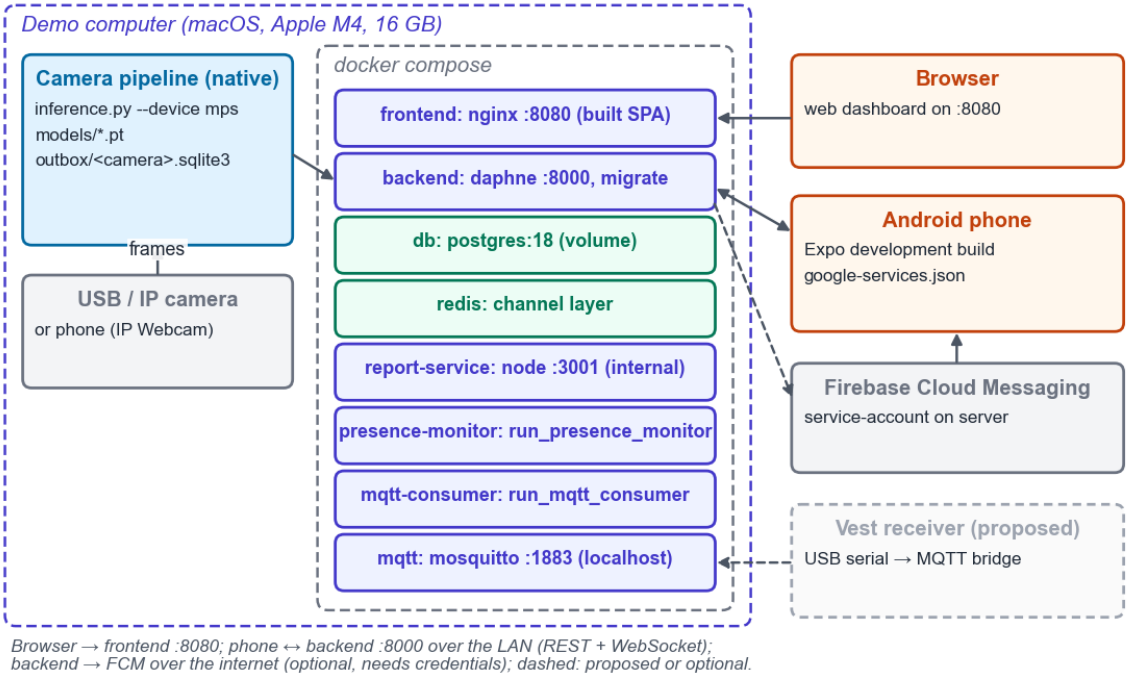



Figure 4.6: Deployment View

Figure 4.7 shows the camera pipeline's structure: main() owns the capture loop and one helper per concern, each free of camera and model imports so that it can be unit-tested without OpenCV, a GPU or weights. The script that draws Figures 4.2 and 4.7 checks every name against the code, so the figures cannot drift from the implementation.

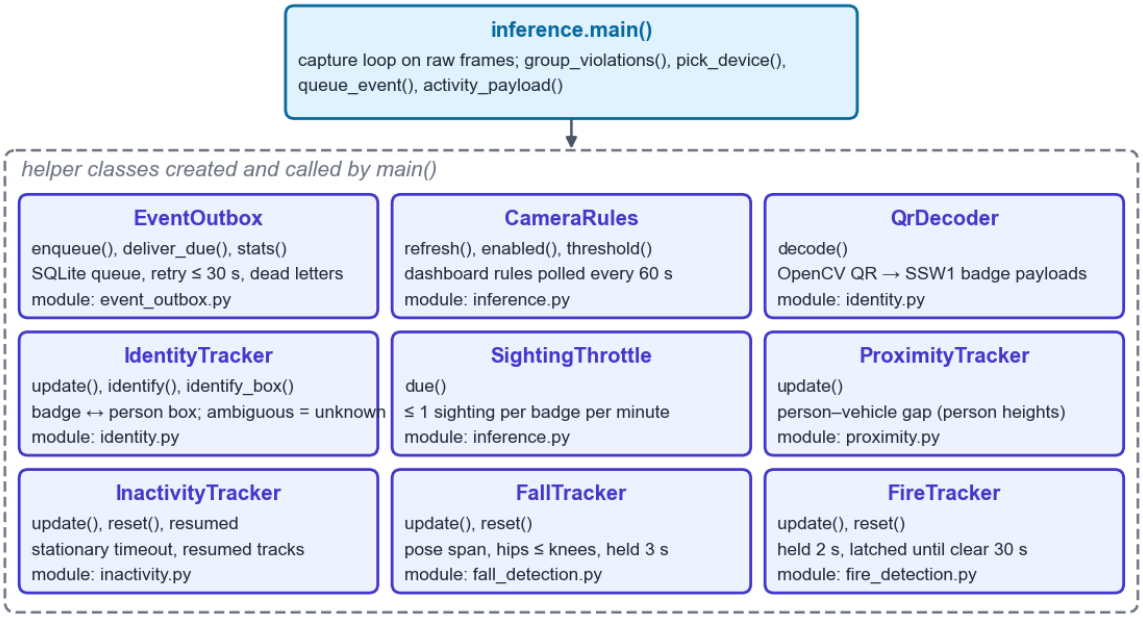



Figure 4.7: Camera Pipeline Class Structure

## 4.9 SOS and Location Telemetry Flow

Figure 4.8 replaces the proposed flow of Figure 3.7 with the implemented sequence. After validation, the handler locks the vest row, updates its position, checks the worker in if needed, finds the highest-risk zone at the position, logs zone entry and raises ZONE_BREACH for an unassigned entry into a restricted zone or for over-occupancy. SOS raises a CRITICAL alert on its rising edge only, so a held button does not repeat it. Changes are broadcast after commit, and the presence monitor later marks a silent vest offline.

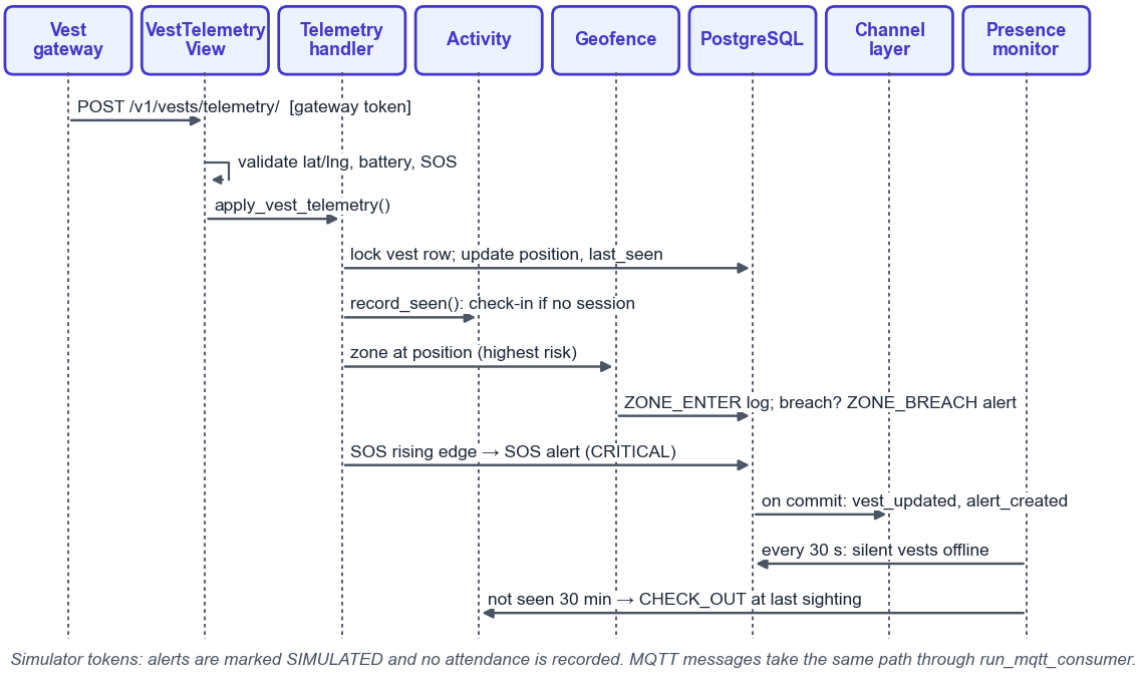



Figure 4.8: Vest Telemetry and SOS Sequence

## 4.10 End-to-End Alert Flow

Figure 4.9 traces a camera alert. The tracker confirms the event, the camera queues and posts it, and the server applies the alert rule and, for fire and smoke, the permit policy before storing it under its event identifier. After commit, the alert is broadcast to web and mobile clients and notifications are sent. The camera deletes the event only after the server replies, so a network failure leaves it queued. Section 5.6 measures this path.

Each AI alert carries its evidence (FR-29): the confidence, the weights that produced it (file name and the first 12 hex digits of its SHA-256, so a retrained model under the same name is told apart) and a JPEG of the unannotated frame that confirmed it, scaled to at most 640 pixels. The snapshot is queued in the outbox with its event and sent in the same request. The server accepts only a JPEG of up to 512 KB, stores it once per event identifier and serves it only to signed-in users. Snapshots can be switched off at the camera or the server, and a daily command deletes them after a retention period (30 days by default) while keeping the alert.

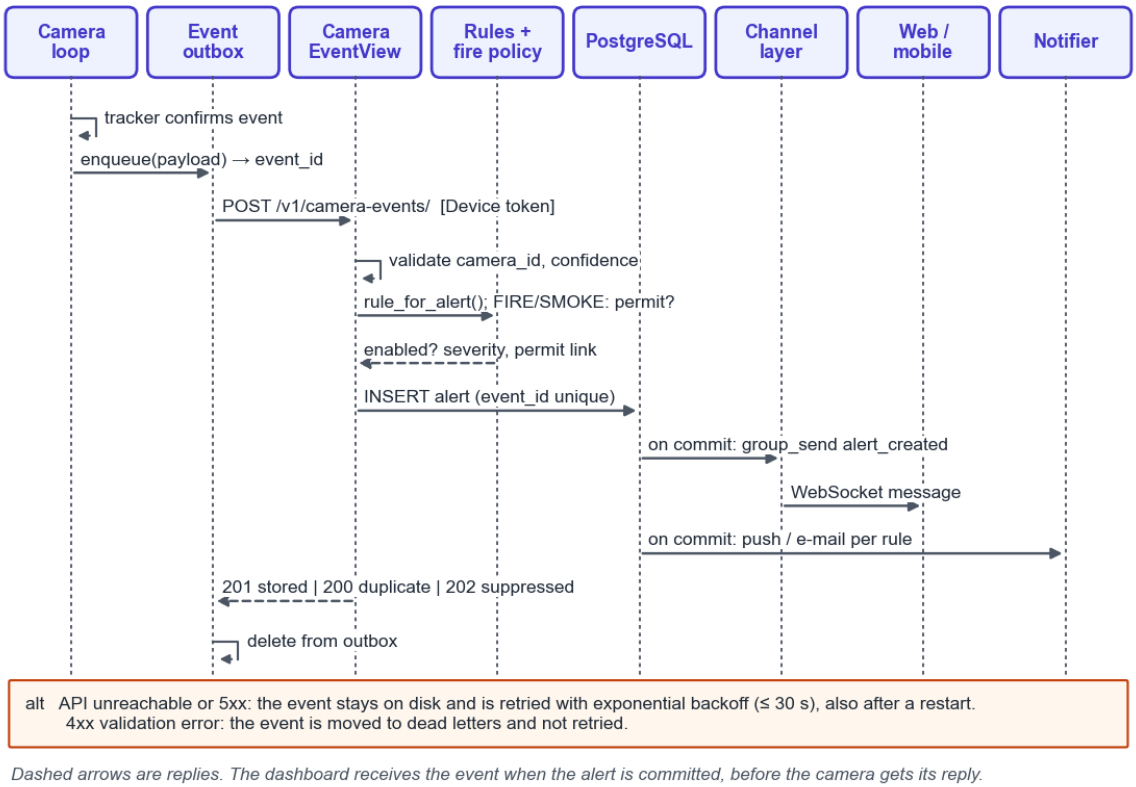



Figure 4.9: Camera Event to Dashboard Sequence

## 4.11 Project Timeline

The final semester followed the completion plan: securing producers and live events first, then replacing fabricated dashboard data and adding worker, zone and breach features, then QR identification, PDF reports, the mobile application and the vest simulator, and finally PostgreSQL verification, containers and CI, fire and smoke with permits, attendance automation and the Chapter 5 measurements. Each stage was recorded with its tests in the requirements traceability matrix.

# Chapter 5

# Testing, Results and Discussion

## 5.1 Testing Approach

Testing was organised around the requirements traceability matrix kept in the repository, which links each functional (FR) and non-functional (NFR) requirement to its code, tests and evidence. A requirement's status rises through five levels, each including the previous one: Coded (the code exists), Tested (automated tests cover the behaviour), Integrated (verified end to end across real running services), Physical (verified with real hardware or a real camera scene) and Accepted (agreed with the supervisor). No requirement has reached Physical or Accepted yet, and the acceptance targets in this chapter are the team's proposals, not targets set by the university.

Four kinds of evidence are used. Automated unit and integration tests run on every change. End-to-end exercises start the real services and drive them with real payloads: camera events, vest telemetry, browser sessions and app sessions. Measurement scripts in tools/measure/ produce the performance and reliability tables; each saves its raw results together with the machine, the git commit and its settings, and a table generator turns them into the tables below, so no number is copied by hand. Model evaluation uses Ultralytics on the held-out test split.

## 5.2 Test Environment

All tests and measurements ran on the development machine: an Apple M4 laptop with 16 GB of memory running macOS (Darwin 27.0), with Python 3.14.3, Node.js 25.8.1 and PostgreSQL 18.3. The backend suite runs against both an isolated SQLite database and PostgreSQL 18, because several defects appeared only on PostgreSQL (Table 5.8). Measurements used a separate PostgreSQL database and simulator device tokens, so their alerts are labelled SIMULATED and never reach report figures. Everything ran on a single host; no measurement yet crosses a real network to a second device.

## 5.3 Automated Test Results

Table 5.1 lists the automated suites. All 318 tests passed when they were re-run for this report on 21 October 2026, up from 59 in the previous revision. Decision logic on the camera side (PPE episodes, fall, inactivity, proximity and fire confirmation) is kept free of OpenCV and model imports, which lets it run under the backend test command without a

camera, a GPU or weights. The web frontend has no unit tests; its lint check and production build are run instead, and its behaviour was checked in the browser (Table 5.7).

Table 5.1: Automated Test Suites

| Suite | Tests | What it covers | Result |
|---|---|---|---|
| Backend (Django) | 183 | Roles and permissions (16); camera and vest ingest, device tokens and alert rules (25); alert evidence: snapshot validation, storage, access and retention (19); fall (24), inactivity (10) and proximity (9) decision logic; WebSocket authentication and broadcasts (8); workers, vests, badges, presence and breaches (33); activity and timelines (18); zones and permits (15); reports (6) | Pass on SQLite and PostgreSQL 18 |
| Camera pipeline | 86 | PPE violation episodes (25), QR identity (13), event outbox with snapshots (11), alert evidence (8), fire confirmation (8), inference configuration (9), activity events (5), dataset audit and split tools (7) | Pass |
| Fire dataset preparation | 4 | Label remapping, video-grouped split, negatives | Pass |
| Report service (Node) | 13 | Metrics, DOCX and PDF rendering, attendance tables, pagination | Pass |
| Mobile app | 16 | API client, token refresh and snapshot fetch (9), push registration (2), state reducers (5) | Pass; lint and type check clean |
| Tools | 16 | Vest simulator (8), measurement statistics and tables (8) | Pass |
| Web frontend | — | No unit tests; ESLint and production build | Lint clean; build OK |
| Total | 318 | | All pass (21 Oct 2026) |

## 5.4 Requirement-Linked Test Cases

Table 5.2 condenses the traceability matrix: each requirement, the evidence gathered for it and the status reached. Statuses are not rounded up. FR-04 (PPE detection) remains Coded although the model runs, because its acceptance criterion asks for event-level precision of at least 90 percent and episode recall of at least 85 percent on a sealed set from the target camera, which does not exist yet. FR-05, one alert per person per violation episode, is Tested: twenty-five tests cover its criterion (two people give two alerts, a one-frame flicker gives none, a continuous violation alerts once), and the full pipeline run headless on a clip with three people missing helmets raised three alerts and no repeats. A 55-second live webcam run then alerted one person up to four times for the same item, because the detector reported one body part as two overlapping boxes; merging such boxes and a 30-second close time fixed this on the recorded box positions (Table 5.8). A second live run still repeated one helmet alert, from a head reported as three boxes, and correctly raised a separate alert for a second person at the edge of the frame; merging nested boxes fixed the

repeat on its recorded boxes. It is not Integrated, because no recorded video with annotated episodes has been scored.

Table 5.2: Requirements, Evidence and Status

| ID | Requirement | Evidence | Status |
|---|---|---|---|
| FR-01 | Sign-in and silent session renewal | Account tests; browser: corrupted token renewed, requests retried | Integrated |
| FR-02 | Roles enforced by the API | 16 permission tests | Tested |
| FR-03 | UI shows only permitted actions | Browser check as Safety Officer and Admin; Viewer not checked in the browser | Integrated |
| FR-04 | Detect missing PPE (10 classes) | Held-out mAP@0.5 0.562 on a leaky split; no sealed target-camera set | Coded |
| FR-05 | One alert per person per violation episode | 25 tests; clip run: 3 people, 3 alerts; two live webcam runs tuned duplicate boxes | Tested |
| FR-06 | Falls from camera pose | 24 tests; no fall video evaluated | Tested |
| FR-07 | Worker-vehicle proximity | 9 tests; still-image trial | Tested |
| FR-08 | Inactivity with dashboard-set timeout | 10 tests; rule polling tests | Tested |
| FR-09 | Fire/smoke with hot-work permits | 20 tests; permit downgrade end to end; model not trained | Integrated (no model) |
| FR-10 | QR worker identification | 20 tests; synthetic-frame end to end | Integrated (synthetic) |
| FR-11 | Reliable camera event delivery | 8 outbox tests; 100-event outage (Table 5.4) | Integrated |
| FR-12 | Only authorised devices submit data | Ingest tests: anonymous, revoked, wrong-kind, wrong-camera tokens | Tested |
| FR-13 | Live updates for every change | Realtime tests; two clients converge; browser | Integrated |
| FR-14 | Authenticated WebSocket (4401) | Realtime tests for missing, bad and expired tokens and origin | Tested |
| FR-15 | Alert rules change runtime behaviour | Ingest tests: disabled type, notify flags, SOS always on | Tested |
| FR-16 | Push notifications | Mocked FCM and app registration tests; no phone yet | Coded |
| FR-17 | E-mail notifications | Tests with a local mail backend | Tested |
| FR-18 | Worker registry in the UI | 7 tests; browser registration with vest | Integrated |
| FR-19 | Zones with polygons and cameras | Zone tests; polygon drawn and saved in the browser | Integrated |
| FR-20 | Automatic breach and over-occupancy alerts | 9 tests; telemetry into a restricted zone end to end | Integrated (simulated GPS) |
| FR-21 | Smart Vest GPS, SOS and link | Simulator scenario end to end; hardware not built | Software integrated; hardware deferred |
| FR-22, 23 | Telemetry validation; vest offline | Validation and presence tests; browser occupancy timeout | Tested; Integrated |
| FR-24 | Activity history: check-in/out, idle | 23 tests; timeline and PDF agree end to end | Integrated |
| FR-25 | Incident list, resolve, export | Resolve broadcast covered by FR-13 | Coded |
| FR-26 | Dashboard shows recorded data only | 3 tests; browser reconciliation | Integrated |
| FR-27 | DOCX and PDF reports | 19 tests; both types and formats end to end | Integrated |
| FR-28 | Mobile app | 13 tests; web build end to end | Integrated on web; no device |
| FR-29 | Alert evidence (snapshot, model version) | 30 tests; outbox to API to web and mobile end to end with a synthetic frame | Integrated (synthetic frame) |

## 5.5 Model Evaluation Results

Table 4.6 gives the held-out results of the deployed YOLO11-medium weights: mAP@0.5 of 0.562, mAP@0.5:0.95 of 0.345, precision 0.467 and recall 0.728. Present-equipment classes exceed 0.95 mAP@0.5, while no-gloves (0.062), no-boots (0.130), gloves (0.276) and no-helmet (0.309) are weak. Because a missing item is the alert condition, the aggregate overstates usefulness for the core task unless the negative classes are read separately.

These figures are an upper bound for two reasons. First, the published split is not independent (Section 4.3.1): 573 source groups and 30 exact-duplicate groups span the training and test splits, so part of the test score measures recall of near-copies. Second, image-level mAP is not alert quality. An alert also depends on the candidate threshold, the grouping of detections per person, persistence and cooldown, none of which has been evaluated on video. A defensible accuracy claim needs the group-constrained re-split and, above all, a sealed set of footage from the target camera, including hard negatives, scored per event.

## 5.6 Performance and Reliability Measurements

Tables 5.3 to 5.6 report the non-functional measurements. Each table states its sample size and settings, and every run is kept, including failures, so a result can be traced to the commit and machine that produced it.

Latency (NFR-01): 100 camera events were posted every 0.2 seconds with a simulator token, and the time from sending each event to its arrival on an authenticated dashboard socket was recorded. All 100 arrived, with a 95th percentile of 41.6 ms against a proposed target of 2 seconds (Table 5.3). The dashboard can receive an event slightly before the camera receives its HTTP response, because the server broadcasts when the alert commits. The figure is for a single host; latency to a phone or a second computer on a LAN has not been measured.

Table 5.3: Event-to-Dashboard Latency (NFR-01)

| Measure | n | p50 (ms) | p95 (ms) | p99 (ms) | Max (ms) |
|---|---|---|---|---|---|
| API response | 100 | 34.2 | 44.0 | 82.2 | 82.3 |
| Event to dashboard | 100 | 32.0 | 41.6 | 80.0 | 81.1 |

Outage recovery (NFR-02): 100 events were sent while the API was unreachable for five minutes, simulated by switching off a proxy, and the camera's sender process was restarted

midway. The target is that every event is stored exactly once within 60 seconds of the API returning. The first run stored all 100 exactly once but took 60.3 seconds, a failure caused by the 60-second retry ceiling; the ceiling was lowered to 30 seconds and two further runs recovered in 30.0 and 30.2 seconds with nothing lost or duplicated (Table 5.4).

Table 5.4: Outage Recovery, 100 Events over a 5-Minute Outage (NFR-02)

| Run (UTC, 20 Oct) | Retry ceiling (s) | Queued at restore | Recovery (s) | Stored once | Lost | Duplicated | Result |
|---|---|---|---|---|---|---|---|
| 11:51 | 60 | 100 | 60.3 | 100 | 0 | 0 | Fail |
| 12:01 | 30 | 100 | 30.0 | 100 | 0 | 0 | Pass |
| 12:13 | 30 | 100 | 30.2 | 100 | 0 | 0 | Pass |

Throughput (NFR-04): the full detector set (PPE every frame; person/vehicle, pose and QR every third frame) was timed on the 517 test images at 1280×720, measuring detection and decoding without capture or display, so camera frame rates are at most these figures. On the laptop GPU the pipeline reached 13.4 and 15.7 FPS against a target of 10; on the CPU it was 9.8 and 10.1 FPS, which led to the GPU becoming the default (Table 5.5). On the GPU the PPE model takes a median of 44.8 ms per frame, and the person/vehicle, pose and QR stages 15–17 ms each on the frames they run. The fire stage is not included because its model is not yet trained, so the figure must be re-measured when it is.

Table 5.5: Inference Throughput with All Available Detectors (NFR-04)

| Device | Run (UTC, 20 Oct) | Frames | FPS | Frame p50 (ms) | Frame p95 (ms) | Result (≥ 10 FPS) |
|---|---|---|---|---|---|---|
| CPU | 11:52 | 150 | 9.8 | 80.4 | 145.5 | Fail |
| CPU | 12:15 | 150 | 10.1 | 82.1 | 147.2 | Pass (marginal) |
| GPU (mps) | 11:51 | 300 | 13.4 | 52.1 | 116.5 | Pass |
| GPU (mps) | 12:14 | 300 | 15.7 | 46.4 | 106.1 | Pass |

Soak (NFR-05): the target is a two-hour run of the full stack with a forced disconnect. Only a five-minute validation run of the script has been made so far (Table 5.6): 146 probes with none missed, no unexpected socket drops and a 1.2-second reconnect after the forced disconnect. API memory rose from 64 to 92 MB in those five minutes, which may be warm-up but cannot be distinguished from a leak without the full run, so NFR-05 is not yet met.

Table 5.6: Soak Validation Run (NFR-05)

| Duration | Probes | Missed | p50 (ms) | p95 (ms) | Socket drops | Forced reconnect (s) | API memory (MB) |
|---|---|---|---|---|---|---|---|
| 5 min (validation) | 146 | 0 | 35.9 | 41.1 | 0 | 1.2 | 63.6 → 91.6 (max 109.8) |

## 5.7 End-to-End Verification

Table 5.7 records the end-to-end exercises, each against real running services. They show that the parts exchange real payloads correctly, but each has a stated boundary: none used a physical camera scene, a phone receiving push or an assembled vest.

Table 5.7: End-to-End Verification Record

| Date | Scenario | Observed result | Boundary |
|---|---|---|---|
| 16 Oct | Token renewal in the browser | Corrupted access token renewed; requests retried with 200 | Single host |
| 16 Oct | Two dashboard clients | Both received create, create and resolve in order | Single host |
| 16 Oct | Outbox replay after outage | 3 events queued: 2 stored, 1 dead-lettered (wrong camera) | Small fixture; see Table 5.4 |
| 16 Oct | Restricted-zone breach | Telemetry into a restricted zone raised a live ZONE_BREACH | GPS simulated over HTTP |
| 16 Oct | QR attribution | Badge decoded; alert attributed to its holder; second person not identified | Synthetic frame; no physical camera |
| 16 Oct | Reports | Both report types in DOCX and PDF through the API; PDF inspected | Test data |
| 16 Oct | Mobile app (web build) | Sign-in, live alert, detail, resolve, cross-session resolve, workers, sign-out | No phone; no push |
| 16 Oct | Vest simulator scenario | Simulated breach and SOS shown; vest offline after silence | Simulator, labelled |
| 20 Oct | Hot-work permit | Fire in permitted zone MEDIUM and linked; elsewhere CRITICAL | Injected events; no fire model |
| 20 Oct | Attendance | 60 min on site, 20 min idle, check-out at last sighting; timeline equals PDF | Synthetic camera events |
| 20 Oct | Alert evidence | Snapshot sent through the outbox; one image stored, a replay stored none; frame and model version shown on web and mobile; retention removed the image and kept the alert | Synthetic frame; no live camera run |
| Earlier | Proximity, real detector | Still image: 4 people, 1 vehicle, CRITICAL alert stored and shown | Still image only |

## 5.8 Defects Found by Testing

Testing and measurement found the defects in Table 5.8 before they reached a demonstration. Three of them appeared only when the backend ran on PostgreSQL instead of the SQLite test database, which is why both now run, and two came from the measurements, which is why each measurement keeps its failed runs.

Table 5.8: Defects Found by Testing

| Defect | Found by | Fix |
|---|---|---|
| Every vest telemetry message failed on PostgreSQL (row lock across a nullable join) | Running the backend tests on PostgreSQL | Lock only the vest row |

| Defect | Found by | Fix |
|---|---|---|
| Outage recovery took 60.3 s, over the 60 s target | Outage measurement (Table 5.4) | Retry ceiling lowered from 60 s to 30 s |
| Full pipeline ran at 9.8 FPS because Ultralytics defaulted to the CPU | Throughput measurement (Table 5.5) | Automatic device selection uses the GPU |
| A late camera event created a second check-in | Activity tests | Late events merged into their session |
| A migration left 100% battery on vests that never reported | Migration check on PostgreSQL | Battery shown as unknown until reported |
| Browsers saw close code 1006, not 4401, for a bad token | WebSocket tests in the browser | Accept, then close with 4401 |
| Report tests were silently not run | Review of test counts | Missing package file added |
| Camera-to-zone lookup used a PostgreSQL-only JSON operator | Tests on SQLite | Portable lookup |
| Fire images whose labels were all invalid became false negatives | Dataset preparation tests | Such images skipped and reported |
| One person got repeat PPE alerts: duplicate same-class boxes became separate episodes | Live webcam run (55 s) | Overlapping or nested same-class boxes merged; episode close time 10 s to 30 s |
| A timeline test failed before noon: its shift was partly in the future | Re-running the suite in the morning | Test uses yesterday's shift |
| The mobile app could not load alert snapshots: the API refused Accept: image/* with 406 | End-to-end check of the mobile web build | The snapshot endpoint ignores the Accept header |

## 5.9 Discussion and Threats to Validity

The software results are strong within their scope. Producers can no longer write anonymously, events survive outages and restarts exactly once, live updates reach clients in tens of milliseconds, and the attendance and report figures reconcile with the stored records. The detection results are weaker. The PPE model is good at confirming that equipment is worn and poor at noticing that it is missing, and the held-out score is inflated by the dataset split, so the system's core detection claim is the least proven part of the project.

Several threats limit how far the results generalise. All measurements ran on one machine, so network effects between camera, server and clients are absent. Camera behaviour was exercised with still images, synthetic frames and injected events rather than footage of a real site, and QR range, fall detection on real falls and fire detection have not been measured at all. The vest path was driven by a simulator. Push delivery to a phone, the two-hour soak and the containerised deployment are untested. The acceptance targets were set by the team. The evidence therefore supports an integrated prototype whose software reliability has been measured and whose detection accuracy still has to be established in the field; it does not support a production-readiness claim.

# Chapter 6

# Conclusion and Future Work

## 6.1 Conclusion

This project set out to replace intermittent manual safety oversight on construction sites with a continuous, camera-first system that identifies the workers involved and keeps an auditable record. The delivered prototype does this end to end in software. Cameras run PPE, person and vehicle, pose and QR models and post events with their own device tokens through an outbox that survives outages. A Django backend applies alert rules and hot-work permits, keeps per-worker attendance and idle time, notifies supervisors and produces DOCX and PDF reports. A React command centre and a React Native mobile application show the results live.

The measurements support the reliability of that software on a single host: a 95th-percentile event-to-dashboard latency of 42 ms, exactly-once delivery of 100 events across a five-minute outage with recovery in about 30 seconds, 15.7 frames per second for the full detector set on the laptop GPU, and 318 passing automated tests. They do not yet support a claim about detection accuracy on a real site. The PPE model recognises missing equipment poorly, its test score is inflated by a leaking dataset split, and fall, fire and QR detection have not been evaluated on real footage. Table 6.1 compares each objective with what was delivered.

## 6.2 Objectives and Outcomes

All ten objectives have working software. Eight are integrated or tested end to end, one (the PPE model) is evaluated offline only, and the physical Smart Vest of objective 8 was moved to future work. The remaining gaps are evaluation with real cameras and devices rather than missing features.

Table 6.1: Objectives and Outcomes

| Objective (Section 1.3) | Delivered | Evidence reached | Outstanding |
|---|---|---|---|
| 1. PPE detection model | YOLO11m, 10 classes, every frame; one alert per person per episode | Offline evaluation, upper bound (mAP@0.5 0.562) | Sealed target-camera set; data for negative classes |
| 2. QR worker identification | Badges, decoding, attribution, sightings | Integrated on synthetic frames | Physical range, angle and lighting test |
| 3. Geofencing | Map polygons; automatic breach and over-occupancy | Integrated with simulated GPS | Real GPS; camera-derived positions |
| 4. Web command centre | Live dashboard, map, permits, reports | Integrated in the browser | Viewer role browser check |
| 5. Mobile application | Alerts, resolve, workers, push registration | Integrated on the web build | Android build and push delivery; rule configuration |

| Objective (Section 1.3) | Delivered | Evidence reached | Outstanding |
|---|---|---|---|
| 6. Workforce activity | Inactivity, automatic check-in/out, idle, timelines | Integrated with synthetic events | Annotated real-camera scenario |
| 7. Incident logging and reports | DOCX and PDF with defined metrics and attendance | Integrated | Supervisor sign-off on the metrics |
| 8. Smart Vest | Telemetry API, MQTT consumer, simulator | Software integrated; hardware moved to future work | Build and field-test the vest |
| 9. Fall detection | Pose geometry with 3 s persistence | Tested | Evaluation on fall video |
| 10. Fire and smoke detection | Confirmation logic and permit policy | Integrated without a trained model | Train the model; evaluate on video |

## 6.3 Limitations

The main limitations follow from the evidence. Detection quality is bounded by a public dataset that does not represent the target cameras, and its split leaks. Proximity and inactivity are measured in pixels on uncalibrated cameras. Worker identity depends on a readable badge and can swap when identified people cross. Notifications are sent from an in-process thread pool, so a crash between storing an alert and sending its notification loses that notification, although the alert itself is kept. The system serves a single site, the site map needs internet access for its tiles, and iOS receives no remote push. All measurements come from one machine.

## 6.4 Future Work

The first priority is the data behind detection. The 69 conflicting label groups should be resolved so the group-constrained split can be used, footage should be collected from the target cameras with hard negatives and the weakest classes, per-class thresholds calibrated, and alerts scored per event and episode on a sealed set, which would also measure the episode tracking of FR-05 on video. The fire and smoke model should be trained on a licence-checked dataset and evaluated on recorded clips, including welding and other hard negatives.

The second priority is physical verification: a QR range and lighting test with printed badges, an Android development build receiving push notifications, the two-hour soak test, latency to a second device over a LAN, a staged fall recording, and a run of the docker-compose deployment and the CI workflow. The measurement scripts already exist, so each of these is a matter of running them on the demonstration setup.

Beyond the FYP, the Smart Vest can be built against the existing gateway interface: an ESP32 with GPS and an SOS button, a point-to-point LoRa link to a USB receiver that

bridges to MQTT, and field tests of range, delivery delay and battery life. Other extensions are a durable queue for notifications, alert-rule configuration in the mobile app, and support for more than one site.

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